



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CONTINUOUS ASSESSMENT TEST - II

WINTER SEMESTER 2023-2024

SLOT: B1 + TB1

Programme Name & Branch : B.Tech
Course Code : BCSE304L
Course Name : Theory of Computation

Faculty Name(s) : Prof. Sathiyakumar, Prof. Anand M, Prof. Lakshmanan K, Prof. Viswanathan P, Prof. Arumuga Arun R, Prof. Shalini L, Prof. Kannadasan R, Prof. Gunavathi C, Prof. Navamani T M, Prof. Rajarajan G, Prof. Madijagan M, Prof. Saritha Murali, Prof. Delhibabu R, Prof. Vishnupriya, Prof. Krishnaraj N, Prof. Bhuvaneswari M, Prof. Kanagaraj R, Prof. Sathya K, Prof. Anand Bihari, Prof. Baskaran P, Prof. Hussain Ahmed Chowdhury

Class Number(s) : VL2023240500758, 0762, 0764, 0767, 0769, 0770, 0773, 0783, 0788, 0794, 0842, 0859, 1011, 1013, 1024, 1027, 1028, 1031, 1034, 1038, 1040

Duration: 90 min.

Max. Marks: 50

Q. No	Question	Marks	Course Outcome (CO)	Bloom's Taxonomy (BL)
1.	a) For languages A and B, let the perfect shuffle of A and B be the language $\{w \mid w = a_1b_1 \cdots a_kb_k, \text{ where } a_1 \cdots a_k \in A \text{ and } b_1 \cdots b_k \in B, \text{ each } a_i, b_i \in \Sigma\}$. Show that the class of regular languages is closed under perfect shuffle.	5	CO3	BL3
	b) Prove using pumping lemma, the following language is not regular $L = \{w \in \{0,1\}^* \mid w \text{ contains more 0's than 1's}\}$	5	CO3	BL2
2.	(a) Design a CFG for the language $L = \{ww^Rzz^R \mid w, z \in \{0,1\}^+, 011 \text{ is a substring of } w \text{ and } z \text{ is odd, } w^R \text{ is the reverse of } w, z^R \text{ is the reverse of } z\}$. Explain the use of each production in the constructed grammar?	5	CO3	BL2
	(b) Convert the given Context free grammar G into an equivalent context free grammar G_1 in Chomsky normal form (CNF) $S \rightarrow aXbX$ $X \rightarrow aY \mid bX \mid \lambda$ $Y \rightarrow X \mid c$	5	CO3	BL2
3.	(a) Show that the language $L = \{\beta\#\beta^R\#\beta \mid \text{where } \Sigma = \{a,c,\#\} \text{ and } \beta \in \{a,c\}^*\}$ is not context free (β^R is the reverse of β).	5	CO3	BL3
	(b) Write the above language L (in Question 3(a)) as the intersection of two context-free languages (over Σ).	5	CO3	BL3



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4.	Consider the following context free grammar (where A is the start symbol) $A \rightarrow BA \mid AC \mid a$ $B \rightarrow CB \mid BA \mid b$ $C \rightarrow AC \mid CB \mid a$ Apply CYK algorithm for the input string abbaa. Find the nonterminal set where X is marked in the table. Without steps full marks will not be awarded. <table border="1"> <tr> <td>X</td><td></td><td></td><td></td><td></td></tr> <tr> <td></td><td>X</td><td></td><td></td><td></td></tr> <tr> <td></td><td>X</td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td>X</td><td></td><td></td></tr> <tr> <td>X</td><td></td><td></td><td></td><td></td></tr> <tr> <td>a</td><td>b</td><td>b</td><td>a</td><td>a</td></tr> </table>	X						X					X						X			X					a	b	b	a	a	10	CO3	BL3
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5.	Give a push down automata (PDA) that recognizes the language L of all strings $w \in \{0,1\}^*$ such that the first and last symbol of w are the same, and moreover, if the length of w is odd, then the middle symbol of w is different from the first and the last. For example, strings 00, 11, 010, 010100010010, 0101010 are in L, but λ, 0, 1, 000, 111, 110, 00001, 1001001 are not. Show the configurations made by the constructed PDA for the input strings (a) 0101010 (b) 111	10	CO4	BL4																														



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CONTINUOUS ASSESSMENT TEST - II

WINTER SEMESTER 2023-2024

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Programme Name & Branch : B.Tech (CSE)
Course Code : BCSE304L
Course Name : Theory of Computation

Duration: 90 min.

Key

Max. Marks: 50

1. (a)

Answer: Let $D_A = (Q_A, \Sigma, \delta_A, q_A, F_A)$ and $D_B = (Q_B, \Sigma, \delta_B, q_B, F_B)$ be two DFAs that recognize A and B , respectively. Here, we shall construct a DFA $D = (Q, \Sigma, \delta, q, F)$ that recognizes the perfect shuffle of A and B .

The key idea is to design D to alternately switch from running D_A and running D_B after each character is read. Therefore, at any time, D needs to keep track of (i) the current states of D_A and D_B and (ii) whether the next character of the input string should be matched in D_A or in D_B . Then, when a character is read, depending on which DFA should match the character, D makes a move in the corresponding DFA accordingly. After the whole string is processed, if both DFAs are in the accept states, the input string is accepted; otherwise, the input string is rejected.

Formally, the DFA D can be defined as follows:

- (a) $Q = Q_A \times Q_B \times \{A, B\}$, which keeps track of all possible current states of D_A and D_B , and which DFA to match.
- (b) $q = (q_A, q_B, A)$, which states that D starts with D_A in q_A , D_B in q_B , and the next character read should be in D_A .
- (c) $F = F_A \times F_B \times \{A\}$, which states that D accepts the string if both D_A and D_B are in accept states, and the next character read should be in D_A (i.e., last character was read in D_B).
- (d) δ is as follows:
 - i. $\delta((x, y, A), a) = (\delta_A(x, a), y, B)$, which states that if current state of D_A is x , the current state of D_B is y , and the next character read is in D_A , then when a is read as the next character, we should change the current state of A to $\delta_A(x, a)$, while the current state of B is not changed, and the next character read will be in D_B .
 - ii. Similarly, $\delta((x, y, B), b) = (x, \delta_B(y, b), A)$.



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1. (b)

The proof is by contradiction. Assume L_4 is regular. Then by the Pumping Lemma, there is an integer $p > 0$, such that for every $w \in L_4$, $|w| \geq p$, there exists a partition of $w = xyz$, such that $|y| > 0$, $|xy| \leq p$, and for all i $xy^iz \in L_4$.

Let $w = 0^{p+1}1^p$ and let x, y, z strings satisfying the conditions of the pumping lemma. Since $|xy| \leq p$ then $y = 0^k$ for some $k \geq 1$. By the Pumping Lemma $w_1 = xy^iz = x(0^k)^iz$ belongs to L_4 for any value of i . But setting $i = 0$ makes the number of 0's in w_1 always equal or less than the number of 1s and $w_1 \notin L_4$. Thus when $i = 0$, w_1 does not belong to L_4 , contradicting the pumping lemma.

2. (a)

$S \rightarrow S_1 S_2$ (S_1 is used for ww^R and S_2 is used for zz^R)

$S_2 \rightarrow 0S_2'0 \mid 1S_2'1 \mid 00 \mid 11$

$S_2' \rightarrow 0S_20 \mid 1S_21$

$S_1 \rightarrow 0S_10 \mid 1S_11 \mid 011 \mid 110$

$X \rightarrow 0X0 \mid 1X1 \mid \epsilon$



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SLOT: B1 + TB1

(b)

Given grammar G

$$S \rightarrow axbx$$

$$x \rightarrow ay | bx | \epsilon$$

$$y \rightarrow x | c$$

After eliminating the null productions

$$S \rightarrow axbx | abx | axb | ab$$

$$x \rightarrow ay | bx | b$$

$$y \rightarrow x | c$$

After eliminating the unit productions.

$$S \rightarrow axbx | abx | axb | ab$$

$$x \rightarrow ay | bx | b \quad y \rightarrow ay | bx | b | c$$

The grammar in CNF

$$S \rightarrow T_1 T_4 | T_2 x | T_3 T_b | T_a T_b$$

$$x \rightarrow T_a y | T_b x | b$$

$$y \rightarrow T_a x | T_b x | b | c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_1 \rightarrow T_a x$$

$$T_2 \rightarrow T_a T_b$$

$$T_3 \rightarrow T_a x$$

$$T_4 \rightarrow T_b x$$



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SLOT: B1 + TB1

3.(a)

Solution Assume that L is context-free and let n be the constant for L prescribed by the stronger version of the pumping lemma. Consider $\alpha := a^n c a^n c a^n \in L$. The pumping lemma gives us the decomposition $\alpha = \beta_1 \beta_2 \beta_3 \beta_4 \beta_5$ with $|\beta_2 \beta_4| \geq 1$ and $|\beta_2 \beta_3 \beta_4| \leq n$. Since $\alpha' := \beta_1 \beta_3 \beta_5 \in L$, $\beta_2 \beta_4$ must not contain the symbol c , i.e., $\beta_2 \beta_4$ consists only of a 's. The condition $|\beta_2 \beta_3 \beta_4| \leq n$ implies that $\beta_2 \beta_4$ can not stretch over all the three runs of a 's in α . Therefore, α' lacks the defining property of the strings of L . This contradiction shows that L is not context-free. •

3.(b)

3b

Define

$$L_1 = \{ \beta \# \beta^r \# \beta \mid \omega, \beta \in \{a, b\}^* \},$$

$$L_2 = \{ \omega \# \beta^r \# \beta \mid \omega, \beta \in \{a, b\}^* \}.$$

Clearly $L = L_1 \cap L_2$. A CFG can be designed which generates L_1 .

$$G_1 = (\Sigma, \{S, U, V\}, S, R)$$

$$S \rightarrow UV$$

$$U \rightarrow \# / a U a / b U b$$

$$V \rightarrow \# / V a / V b.$$

A similar grammar can be given for L_2 .



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CONTINUOUS ASSESSMENT TEST - II

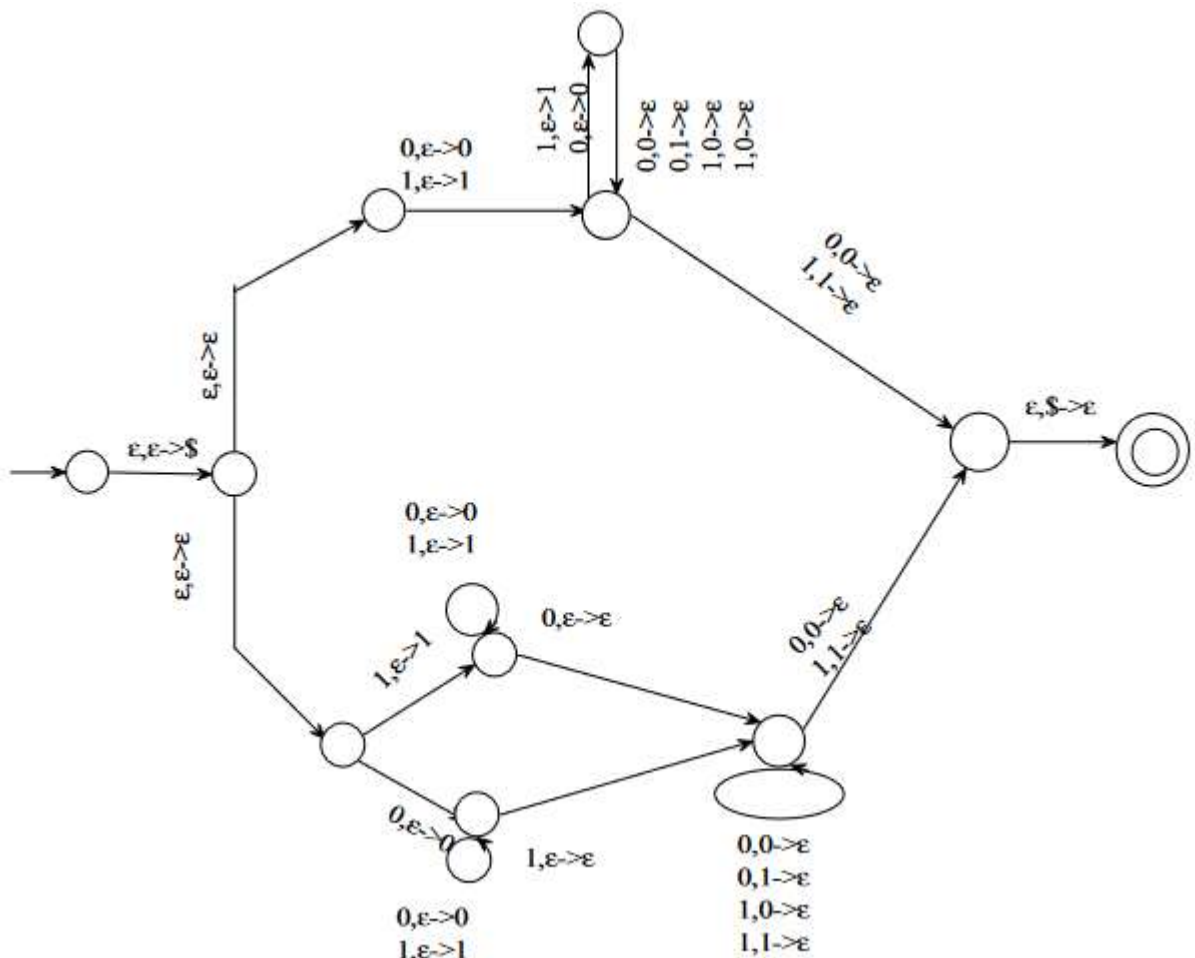
WINTER SEMESTER 2023-2024

SLOT: B1 + TB1

4. Explanation should be given for the CYK algorithm with the necessary steps

A,B,C				
A,B,C	A,B,C			
B,C	A,B	A,B,C		
B,C	ϕ	A,B	A,C	
A,C	B	B	A,C	A,C
a	b	b	a	a

- 5.





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SLOT: B1 + TB1

The PDA recognizes the language by guessing the length of the input at the beginning. For the guess that the length is even it follows the upper branch, while for the guess that length is odd it follows the lower branch.

- 1) In the upper branch, the PDA accepts even length strings by pushing and popping alternatively, starting with the second symbol read and ending before the last symbol.
- 2) In the lower branch, the PDA first pushes the symbols onto the stack and guesses the middle symbol. If all the symbols except the first one and the start symbol \$ can be popped out when the PDA finishes reading the symbol except the last one, then the guess is right. The lower branch remembers the first symbol by developing two branches, Each branch accepts when the guessed middle symbol is different from the first one.
- 3) Before entering the accept state, the PDA only pops out the first read symbol out of the stack when the last one read is the same. Thus it only accepts the string starting and ending with the same symbol.



School of Computer Science and Engineering

Winter Semester 2023-24

Continuous Assessment Test – II

SLOT: B2+TB2

Programme Name & Branch: B.Tech

Course Name & Code: Theory of Computation & BCSE304L

Class Number (s): VL2023240500789, VL2023240501047, VL2023240501029, VL2023240500774, VL2023240500845, VL2023240500760, VL2023240501015, VL2023240500784, VL2023240501045, VL2023240500861, VL2023240501036, VL2023240500763, VL2023240501026, VL2023240500766, VL2023240500771, VL2023240501043, VL2023240500797, VL2023240501049, VL2023240501037, VL2023240501042, VL2023240501051, VL2023240501014.

Faculty Name (s): NAVAMANI T M, KONATHAM, SUMALATHA, KANAGARAJ R, KANNADASAN R, MADIAJAGAN M, SATHIYA KUMAR C, MOHANA CM, KRISHNA RANI, SAMAL K, UMAMAHESWARI M, SARITHA MURALI, ANAND BIHARI, ANAND M, KRISHNARAJ N, ARUMUGA, ARUN R, SHALINI L, SARWESH P, RAJARAJAN G, SABYASACHI KAMILA, SOMASUNDARAM S K, HUSSAIN AHMED CHOWDHURY, UMA PRIYA D, VISHNUPRIYA.

Exam Duration: 90 Min.

Maximum Marks: 50

Q. No.	Question	Max Marks	CO	BL
1.	a) The derivative of a language for a string x is defined as follows: $L_x = \{ y \mid xy \in L \}$. 1) what is L_{aa} for the language $L = \{ a^{n+1}b^n \mid n \geq 1 \}$. (1mark) 2) what is L_b for the language $L = \{ a^{n+1}b^n \mid n \geq 1 \}$. (1 mark) 3) Prove that when L is regular, the derivative of L (ie., L_x) is also regular. (3 marks)	5	CO3	BL3
	b) Consider the language $L_1 = \{ a^p b^q a^r \mid p, q, r \geq 1 \text{ and } p+q \neq r \}$. Prove or disprove that the language L is regular.	5	CO3	BL2
2.	a) Design a CFG for the language $\gamma = \{ a^n b^m a^{2n} \mid n, m \geq 0 \}$.	5	CO3	BL2
	b) Consider the grammar $S \rightarrow 0S1 \mid 1S0 \mid \lambda$, where S is the start symbol, $\{0,1\}$ are terminals. You can see that for every 0 there is a 1 and vice versa in the rule (other than $S \rightarrow \lambda$). Is the grammar generates the language $L_2 = \{ w \mid w _0 = w _1 \}$ where $ w _a$ refers to the occurrences of a in w . If not, justify with a string in the language L_2 that cannot be produced by the given grammar rules. What further changes you need to do in the given grammar in order to generate L_2 .	5	CO3	BL2
3.	a) Construct the CYK algorithm for the following CFG and check whether the string “aabc b” is derivable from the grammar. The Start Symbol is A.	10	CO3	BL3

	$A \rightarrow CB$ $B \rightarrow DC \mid b$ $C \rightarrow BD \mid a$ $D \rightarrow AD \mid c$			
4.	a) Construct the given Context Free Grammar $G=(V,T,P,S)$ into Chomsky normal form. P is the start symbol. $P \rightarrow RPT$ $Q \rightarrow RP/Ra$ $R \rightarrow bRPR \mid b \mid \epsilon$ $T \rightarrow PaP \mid R \mid aa$	5	CO3	BL3
	b) Derive the CFG for the given Context Free Language $L = \{v^r i^{r+s} t^s \mid r, s \geq 0\}$ and check the ambiguity of the given grammar.	5	CO3	BL3
5.	The language L have a strings of balanced “#” symbol and “@” symbol : every “#” symbol can be paired with a unique subsequent of “@” symbol , and every “@” symbol can be paired with a unique preceding “#” symbol. Moreover, the string between any such pair has the same property. Derive the language for the given condition and Construct the PDA for the derived language.	10	CO4	BL4

CAT 2 — B2 + TB2.

1. a) i) $L = \{y \mid xy \in L\} \quad L = \{aab, aabbb, aaaaa bbb, \dots\}$
 $Laa = \{a^{n+1}b^n \mid n \geq 1\} \quad Laa = \{b \mid aab \in L\}$

$$Laa = \{aa \rightarrow \underbrace{a^{n-1}b^n}_y\}$$

$$= \{a^{n-1}b^n \mid n \geq 1\}$$

$$Laa = \{b^n, ab^n, a^2b^n, a^3b^n, \dots\} \checkmark$$

(ii) $L_b = \{y \mid by \in L\} \quad xy \in L$

$$L_b = \emptyset$$

because The string does not start with b.

(iii) if L is regular Then the derivative also regular

$$L_a = \{a^n b^m \mid n \geq 1, m \geq 1\}$$

it is possible to construct FA.

1. b

1. b. $L_1 = \{a^p b^q a^r \mid p, q, r \geq 1 \text{ and } p+q \neq r\}$ is Regular?

There is no mechanism to keep record of $p+q$ is whether equal or not equal to r .

Therefore it is not possible to implement by an automata. Therefore given language is not regular.

2) a) $Y = \{a^n b^m c^{2n} \mid n, m \geq 0\}$ Give CFG.

$n=0, m=0 \quad \epsilon$.

$n=1, m=0 \quad a \quad cc$

$n=0, m=1 \quad b$

$n=1, m=1 \quad a bcc$.

$S \rightarrow aScc \mid \epsilon$ ~~$a^n b^m c^{2n}$~~ A .

$A \rightarrow bA \mid \epsilon$.

Derivation: $w = aabccccc$

$S \rightarrow aScc$

$\rightarrow aaSccc$

$\rightarrow aaAcccc$

$\rightarrow aabAcccc$

$\rightarrow aab\epsilon cccc$

$\rightarrow aabccccc$.

2) b)

3) $A \rightarrow CB$
 $B \rightarrow DC/b$
 $C \rightarrow BD/a$
 $D \rightarrow AD/c$ $w = aabcb$

Given grammar is in CNF.

5	ϕ				
4	ϕ	ϕ			
3	ϕ	D	A		
2	ϕ	A	C	ϕ	
1	C	C	B	D	B
	a	a	b	c	b

$$\hat{i}=1, \hat{j}=3, k=1$$

$$V_{13} = V_{11} V_{22}$$

$$C \quad A$$

$$CA$$

$$= \phi$$

$$\hat{i}=1, \hat{j}=3, k=2$$

$$V_{13} = V_{12} V_{31}$$

$$\phi \quad B$$

$$= B$$

$$= \phi$$

$$\hat{i}=2, \hat{j}=3, k=1$$

$$V_{23} = V_{21} V_{22}$$

$$= C \quad A$$

$$= CA$$

$$= \phi$$

$$\hat{i}=2, \hat{j}=3, k=2$$

$$V_{23} = V_{22} V_{31}$$

$$= A \quad D$$

$$= AD \in D$$

$$= D$$

$w = aabcb$ is not belongs to the given grammar.

$$4) \text{ a) } G: P \rightarrow RPT$$

$$Q \rightarrow RP/Ra$$

$$R \rightarrow bRPR/b/\epsilon$$

$$T \rightarrow PaP/R/aa.$$

(i) Removal of useless symbol Q since it is not reachable.

$$P \rightarrow RPT$$

~~$$Q \rightarrow RP/Ra$$~~

$$R \rightarrow bRPR/b/\epsilon$$

$$T \rightarrow PaP/R/aa.$$

(ii) Removal of $R \rightarrow \epsilon$.

$$P \rightarrow RPT/PT$$

$$R \rightarrow bRPR/bRP/bPR/bP/b$$

$$T \rightarrow PaP/R/\epsilon/aa.$$

Removal of $T \rightarrow \epsilon$

$$P \rightarrow RPT/RP/PT/P$$

$$R \rightarrow bRPR/bRP/bPR/bP/b$$

$$T \rightarrow PaP/R/aa.$$

(iii) Removal of unit productions $P \rightarrow P, T \rightarrow R$

$$P \rightarrow RPT/RP/PT$$

$$R \rightarrow bRPR/bRP/bPR/bP/b$$

$$T \rightarrow PaP/bRPR/bRP/bPR/bP/b/aa.$$

(iv) Reducing length.

$$P \rightarrow R A / R P / P T$$

$$R \rightarrow b B / b D / b C / b P / b$$

$$T \rightarrow P E / b B / b D / b C / b P / b / a a$$

$$A \rightarrow P T$$

$$B \rightarrow R C$$

$$C \rightarrow P R$$

$$D \rightarrow R P$$

$$E \rightarrow a P$$

(v) Replacing terminal by introducing new NT

$$P \rightarrow R A / R P / P T$$

$$R \rightarrow G B / G D / G C / G P / b$$

$$T \rightarrow P E / G B / G D / G C / G P / b / F F$$

$$A \rightarrow P T$$

$$B \rightarrow R C$$

$$C \rightarrow P R$$

$$D \rightarrow R P$$

$$E \rightarrow F P$$

$$F \rightarrow a$$

$$G \rightarrow b$$

4) b) $L = \{ v^r i^{r+s} t^s \mid r, s \geq 0 \}$ Ambiguity.

Grammar is

$$L = v^r i^{r+s} t^s$$

$$S \rightarrow xy$$

$$x \rightarrow vx i / \epsilon$$

$$y \rightarrow i y t / \epsilon.$$

This

LMD

$$S \rightarrow xy$$

$$\rightarrow vx i y$$

$$\rightarrow vvx i i y$$

$$\rightarrow vv i i y$$

$$\rightarrow vv i i i y t$$

$$\rightarrow vv i i i t \epsilon L.$$

$$S \rightarrow xy$$

No way to derive.

(5)

@ # # # @ # @ @ # (@ @)

@, @ | @ @

@, z0 | @ z0

#, z0 | # z0

#, # | # #

#, @ | ε

@, # | ε

@, # | ε

#, @ | ε

#, z0 | # z0

#, # | # #

@, @ | @ @

@, z0 | @ z0

(08) #, # | # #

@, @ | @ @

@, z0 | @ z0

#, z0 | # z0

ε, z0 | z0

@, # | ε

#, @ | ε



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

Programme Name & Branch : B.Tech, CSE
Course Code and Course Name : BCSE304L – Theory of Computation
Faculty Name(s) : KANAGARAJ R, MADIAJAGAN M, SATHIYA KUMAR C, KARTHIK G M, BASKARAN P, PARTHASARATHY G, SARITHA MURALI, RAJARAJAN G , SHALINI L, UMA PRIYA D, BOOMINATHAN P, LAKSHMANAN K, BHAWANA TYAGI, BHUVANESWARI M, IYAPPAN P, ISLABUDEEN M, PRAKASH M, SATHYA K, ADRIJA BHATTACHARYA, DEBI PRASANNA ACHARJYA, K.KRISHNA RANI SAMAL, SUGANTHINI C
Class Number(s) : VL2024250501633, 1615, 1619, 1625, 1637, 1639, 1635, 1653, 1631, 1643, 1617, 1651, 1649, 1645, 1627, 1647, 1629, 1641, 1623, 1613, 1621, 1655
Date of Examination : 21-Mar-2025, 2:00 PM – 3.30 PM
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

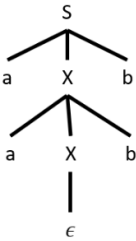
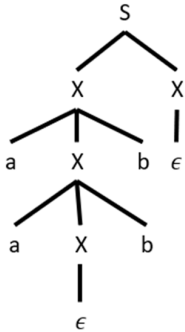
- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Q. No	Question	M	CO	BL
1.	a) Prove that the given language $L = \{a^n b^m \mid 1 \leq n < m\}$ is regular language or not. Ans: If L is regular, it satisfies the Pumping Lemma for pumping length p such that $w \in L$ with $w \geq p$. Split w into xyz, where 1. $xy^i z \in L, \forall i \geq 0$ 2. $y > 0$ 3. $xy \leq p$. $L = \{a^n b^m \mid 1 \leq n < m\}, w = a^l b^l$, i.e $w = p$ Choose P = 5, so $w = aabbbb \in L$. Split into xyz for w such that $x = a, y = ab, z = bb$ 1. $y = ab = 2 > 0$, TRUE 2. $xy = aab = 3 \leq (p = 5)$, TRUE 3. Check $xy^i z \in L$ for $i=1, 2, 3, 4, 5...$ $i=2: xy^2 z \rightarrow a(ab)^2 bb = aababbb \notin L$ Therefor the given language is not Regular.	7		
	b) Consider the homomorphism h from the alphabet $\{0,1,2\}$ to $\{a,b\}$ defined by: $h(0)=ab, h(1)=b, h(2)=aa$. 1. What is $h(0210)$ and $h(2201)$? 2 marks 2. If L is the language $(ab+baa)^*bab$, what is $h^{-1}(L)$? 1 mark Ans: 1. $h(0210) = abaabab, h(2201) = aaaaabb$ 2. $h^{-1}(L) = (0 + 12)^*10$	3		



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SLOT: F1+TF1

2.	a) Consider the grammar $X \rightarrow aXb \mid XX \mid \epsilon$. Determine whether the grammar is ambiguous. If it is, prove the ambiguity by deriving two distinct parse trees for the same string. Ans: For String aabb lm $S \Rightarrow aXb$ lm $\Rightarrow aaXbb$ lm $\Rightarrow aa\epsilon bb$ lm $\Rightarrow aabb$ rm $S \Rightarrow XX$ rm $\Rightarrow aXbX$ rm $\Rightarrow aXb\epsilon$ rm $\Rightarrow aaXbb$ rm $\Rightarrow aa\epsilon bb$ rm $\Rightarrow aabb$  	3		
	b) Design context free grammar for the language $L = \{a^i b^j c^k \mid i = j + k\}$ Ans: $L = \{\epsilon, a, ac, ab, aabb, aacc, aabc, aaabbb, aaabbc, aaabcc, aaaccc, \dots\}$ $S \rightarrow AB,$ $A \rightarrow aAb \mid aAc \mid \epsilon,$ $B \rightarrow bB \mid cB \mid \epsilon$ String $a^4 b^3 c$ $S \Rightarrow AB$ $\Rightarrow aAcB$ $\Rightarrow aaAbcB$ $\Rightarrow aaaAbbbcB$ $\Rightarrow aaaaAbbbcbB$ $\Rightarrow aaaa\epsilon bbbcbB$ $\Rightarrow aaaaabbbcb\epsilon$ $\Rightarrow aaaaabbbcb$ String accepted	4	3	2
	c) Consider the grammar G with $S \rightarrow 0S1 \mid 1S0 \mid \epsilon$. In this grammar, for every 0 there is a 1 and vice versa. Is this grammar generating $L = \{w \mid n_0(w) = n_1(w)\}$? Justify. If not, what further modifications are required in the grammar in order to generate the language L? Ans: Examples of strings generated by G: - xx 0x10x1	3		



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

	• $1x01x0$ • $0(0x1)10(0x1)1 \rightarrow 00x1100x11$ • $1(0x1)01(0x1)0 \rightarrow 10x01$ The justification is as follows: 1. Each production rule ensures that for every 0, there is a matching 1, and vice versa. 2. The recursive structure ensures that the balance is maintained at every step of the derivation. 3. The base case x provides a single valid terminal string, ensuring $n_a(w)=n_b(w)=0$ (a valid element of L). Base Case: For $S \rightarrow x$, the string x has no 0s or 1s, so $n_a(w)=n_b(w)=0$, which satisfies the condition. Inductive Step: Assume that a string ww derived from S satisfies $n_a(w)=n_b(w)$. When applying $S \rightarrow 0S1$ or $S \rightarrow 1S0$, the added 0 and 1 (or 1 and 0) maintain the balance. Therefore, any string derived from SS satisfies $n_a(w)=n_b(w)$. Hence, the grammar is correct for generating L.			
3.	a) Consider the grammar $S \rightarrow ASA \mid aB, A \rightarrow B, B \rightarrow b \mid \epsilon$. Convert into Chomsky Normal Form Ans: 1. S appears in RHS, $S \rightarrow ASA$ so $P: \{ S' \rightarrow S, S \rightarrow ASA \mid aB, A \rightarrow B, B \rightarrow b \mid \epsilon \}$ 2. Remove ϵ- production. $B \rightarrow \epsilon : P: \{ S' \rightarrow S, S \rightarrow ASA \mid aB \mid a, A \rightarrow B \mid \epsilon, B \rightarrow b \}$ $A \rightarrow \epsilon : P: \{ S' \rightarrow S, S \rightarrow ASA \mid AS \mid SA \mid S \mid aB \mid a, A \rightarrow B, B \rightarrow b \}$ 3. Remove unit production $S \rightarrow S : \text{Recursive so remove from } P: \{ S' \rightarrow S, S \rightarrow ASA \mid AS \mid SA \mid aB \mid a, A \rightarrow B, B \rightarrow b \}$ $A \rightarrow B : P: \{ S' \rightarrow S, S \rightarrow ASA \mid AS \mid SA \mid aB \mid a, A \rightarrow b, B \rightarrow b \}$ $S' \rightarrow S : P: \{ S' \rightarrow ASA \mid AS \mid SA \mid aB \mid a, S \rightarrow ASA \mid AS \mid SA \mid aB \mid a, A \rightarrow b, B \rightarrow b \}$ 4. Production more than 2 not terminals in RHS $S' \rightarrow ASA : S' \rightarrow AX, X \rightarrow SA$, Similarly for $S \rightarrow ASA$. So $P: \{ S' \rightarrow AX \mid AS \mid SA \mid aB \mid a,$ $S \rightarrow AX \mid AS \mid SA \mid aB \mid a,$ $A \rightarrow b,$ $X \rightarrow SA,$ $B \rightarrow b \}$ 5. RHS should have only one Terminal $S' \rightarrow aB : S' \rightarrow YB, Y \rightarrow a$. So $P: \{ S' \rightarrow AX \mid AS \mid SA \mid YB \mid a,$ $S \rightarrow AX \mid AS \mid SA \mid YB \mid a,$ $A \rightarrow b,$ $X \rightarrow SA,$ $Y \rightarrow b$ $B \rightarrow b \}$ 6. Hence given grammar is in CNF	6	3	4
	b) Consider the context-free grammar whose rules are $S \rightarrow SS \mid Sa \mid a$. Convert into equivalent grammar in GNF.	4		



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

	Ans: $S \rightarrow SS Sa a$ 1. Remove Left Recursion $S \rightarrow SS$ substitute $S \rightarrow a$ then $S \rightarrow aS$ $P: \{S \rightarrow aS, S \rightarrow Sa, S \rightarrow a\}$ 2. Remove Left Recursion $S \rightarrow Sa$ substitute $S \rightarrow a$ $P: \{S \rightarrow aS, S \rightarrow aa, S \rightarrow a\}$ Given grammar has RHS starts with a terminal. So it is now in GNF.																																							
4.	Given the context-free grammar $G: S \rightarrow XY, X \rightarrow YY \mid 0, Y \rightarrow XY \mid 1$. Use the CYK algorithm to determine whether the strings 00110 is in the language generated by the given grammar. Ans: <table border="1"> <tr><td>-</td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>-</td><td>-</td><td></td><td></td><td></td><td></td></tr> <tr><td>-</td><td>-</td><td>-</td><td></td><td></td><td></td></tr> <tr><td>-</td><td>S</td><td>X</td><td>-</td><td></td><td></td></tr> <tr><td>X</td><td>X</td><td>Y</td><td>Y</td><td>X</td><td></td></tr> <tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td></td></tr> </table> String is not accepted.	-						-	-					-	-	-				-	S	X	-			X	X	Y	Y	X		0	0	1	1	0		10	3	3
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0	0	1	1	0																																				
5.	a) Consider the PDA for accepting palindrome strings over $\{a, b\}$. For the below incomplete trace sequence (instantaneous description – ID) and moves for the given string ababababa, identify the required state or partial input string or stack content in the “fill in the blanks” entry. Note q_0 is the start state and q_f is the final state, Z_0 is the start stack symbol. $(q_0, ababababa, Z_0)$ $\vdash (q_0, babababa, aZ_0)$ $\vdash (q_0, abababa, baZ_0)$ $\vdash (q_0, bababa, -)$ $\vdash (q_0, ababa, -)$ $\vdash (-, baba, babaZ_0)$ $\vdash (q_1, -, abaZ_0)$ $\vdash (q_1, ba, baZ_0)$ $\vdash (q_1, a, aZ_0)$ $\vdash (-, \epsilon, Z_0)$ $\vdash (q_f, \epsilon, Z_0)$ Ans:	5	1	4																																				



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

	$a, Z_0 \mid aZ_0$ $b, Z_0 \mid bZ_0$ $a, a \mid aa$ $b, b \mid bb$ $b, a \mid ba$ $a, b \mid ab$ Trace "ababababa": $\begin{aligned} &\delta(q_0, ababababa, Z_0) \vdash (q_0, babababa, aZ_0) \\ &\vdash (q_0, abababa, baZ_0) \\ &\vdash (q_0, bababa, abaZ_0) \\ &\vdash (q_0, ababa, babaZ_0) \\ &\vdash (q_1, baba, babaZ_0) \\ &\vdash (q_1, aba, abaZ_0) \\ &\vdash (q_1, ba, baZ_0) \\ &\vdash (q_1, a, aZ_0) \\ &\vdash (q_1, \epsilon, Z_0) \\ &\vdash (q_2, \epsilon) \end{aligned}$ String is accepted			
	b) Consider the language $L = \{a^{2n}c^mb^n \mid n, m \geq 1\}$. The objective is to construct a (pushdown automata) PDA for this language and below are the partial transition rules. The main idea is to push a 0 in the stack whenever a is read and when bs are read, they need to be cancelled against 0s such a way that for every one b is read, two 0s need to be cancelled out in the stack. You are expected to fill-in the remaining rules which are left un-attempted (denoted with - in the rules) and complete the rules such that the constructed PDA will accept the given language L. FYI, the transition function δ of the PDA is defined as: $\delta : Q \times \Sigma \cup \{\lambda\} \times \Gamma \rightarrow \text{finite subset of } Q \times \Gamma^*$. You are allowed to use only one (new) additional state which is q_f, the final state. Write only incomplete rules 2,4,6,7,8 for answering the question. Z is the start stack symbol. a. $\delta(q_0, a, Z) = (q_1, 0Z)$ b. $\delta(q_1, a, -) = (q_1, 00)$	5		



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
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	Ans: c. $\delta(q1, c, 0) = (q2, 0)$ d. $\delta(q2, c, -) = (-, -)$ e. $\delta(q2, b, 0) = (q3, \lambda)$ f. $\delta(q3, -, -) = (-, \lambda)$ g. $\delta(q4, b, -) = (-, -)$ a. $\delta(q4, \lambda, -) = (-, -)$ a) $\delta(q0, a, z) = (q1, 0z)$ b) $\delta(q1, a, 0) = (q1, 00)$ c) $\delta(q1, c, 0) = (q2, 0)$ d) $\delta(q2, c, 0) = (q2, 0)$ e) $\delta(q2, b, 0) = (q3, \lambda)$ f) $\delta(q3, \lambda, 0) = (q4, \lambda)$ g) $\delta(q4, b, 0) = (q3, \lambda)$ h) $\delta(q4, \lambda, z) = (qf, z)$			
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F2+TF2

Programme Name & Branch : B.Tech CSE and Specialization
Course Code and Course Name : BCSE304L Theory of Computation
Faculty Name(s) : Prof.Kannadasan R, Prof. Mohana C M , Prof. Annapurna Jonnalagadda, Prof. Navamani T M , Prof. Arpan Garai, Prof. Anand Bihari , Prof. Uma Priya D , Prof. Saravanan R , Prof. Naga Priyadarsini R , Prof. Krishnaraj N , Prof. Gunavathi C , Prof. Viswanathan P, Prof. Vijayarajan V, Prof. Vishnu Priya A , Prof. Sridevi S , Prof. Prabakaran N, Prof. Sri Preethaa K R , Prof. Katari Balakrishna
Class Number(s) : VL2024250501656, 1634, 1622, 1628, 1648, 1632, 1644, 1640, 1654, 1642, 1618, 1620, 1616, 1638, 1636, 1624, 1650, 1614
Date of Examination : 21.03.2025 FN
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes(CO):
 CO2: Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
 CO3: Identify limitations of some computational models and possible methods of proving them.
 CO4: Represent the abstract concepts mathematically with notations.

Q. No	Question	M	CO	BL
1.	(a) Convert the following Grammar into Finite automata $S \rightarrow aA bB$ $A \rightarrow aA bS c$ $B \rightarrow bB a$ (3 Marks) (b) Let R_1 and R_2 be regular languages over Σ . Then, exclusive of R_1 and R_2 is defined as $EXC(R_1, R_2) = \{ x \in \Sigma^* : x \text{ in } \{R_1 - R_2\} \text{ or } x \text{ in } \{R_2 - R_1\} \}$. If $x \text{ in } \{R_i - R_j\}$ means x is in R_i and not in R_j . Prove that $EXC(R_1, R_2)$ is regular. (3 Marks) (c) Using pumping lemma, demonstrate that the given language $L = \{x^i y^j z^k \mid k \leq i+j\}$ is not regular. (4 Marks)	10	CO 2	BL2



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SLOT: F2+TF2

	1 a) b) $EXC(R_1, R_2) = \{x \in \Sigma^+ \mid x \in (R_1 - R_2) \text{ or } x \in (R_2 - R_1)\}$ $(R_1 - R_2) = \{x \in R_1, x \notin R_2\}$ $(R_2 - R_1) = \{x \notin R_1, x \in R_2\}$ Regular language closed under → Intersection → union → Complement → Set difference. We use the closure properties to prove $EXC(R_1, R_2)$ is regular.			
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SLOT: F2+TF2

Step 1:- $R_1 - R_2$ is the set of strings in R_1
but not in R_2 .

$R_1 \cap \bar{R}_2$ where $\bar{R}_2$ is the complement of R_2

∴ since regular language are closed under
complement and intersection $R_1 - R_2$ is
regular.

Step 2:-

$R_2 - R_1$ is the set of strings in R_2
but not in R_1

$R_2 \cap \bar{R}_1$ $\bar{R}_1$ is the complement of R_1

Similarly $R_2 - R_1$ is regular.

Steps:-

$$\text{EXC}(R_1, R_2) = (R_1 - R_2) \cup (R_2 - R_1)$$

$R_1 - R_2$ and $R_2 - R_1$ is regular
under Union



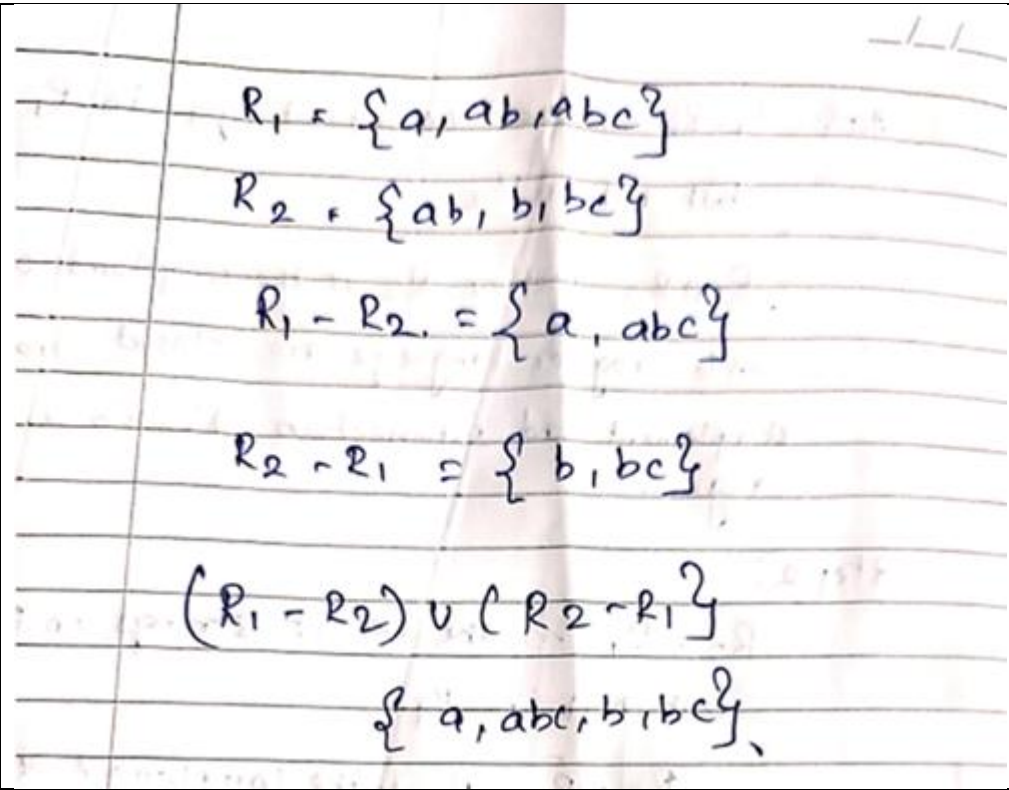
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SLOT: F2+TF2

 $R_1 = \{a, ab, abc\}$ $R_2 = \{ab, b, bc\}$ $R_1 - R_2 = \{a, abc\}$ $R_2 - R_1 = \{b, bc\}$ $(R_1 - R_2) \cup (R_2 - R_1)$ $\{a, abc, b, bc\}$			
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WINTER SEMESTER 2024-2025

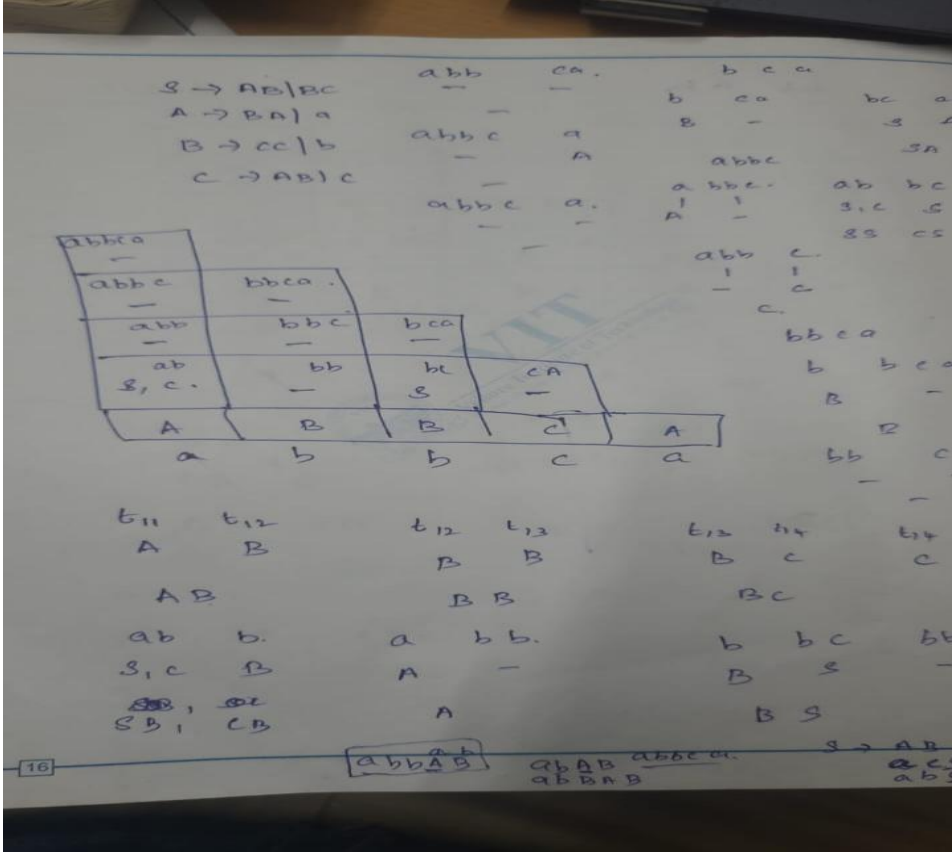
SLOT: F2+TF2

	Turning machine 1) $x^i y^j z^k$ $k \leq i+j$ $i=2$ $j=3$ $k=5$ $x^2 j^3 z^5$ $xxjjjzzzzz$ $S \geq m$ $S = uv^i w$ $uv \leq m$ $v \geq 1$ $\frac{S}{w} = uv^i w$ $\frac{xxjjjzzzzz}{u \quad v \quad w}$ $i=2$ $xxjjjjzzzzz$ is not part of the language.			
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F2+TF2

2.	Consider the following context free grammar (where S is the start symbol) $S \rightarrow AB \mid BC$ $A \rightarrow BA \mid a$ $B \rightarrow CC \mid b$ $C \rightarrow AB \mid c$ Apply CYK algorithm with all required steps for the input string "abbca". Answer : 	10	CO 3	BL3
3.	a) Consider the following grammar (where U is the start symbol) remove all unit-productions, all useless productions, and all λ - productions from the grammar where $\Sigma = \{0,1\}$ $U \rightarrow 0V \mid 0WW,$ $V \rightarrow 0V \mid \lambda,$ $W \rightarrow 1W \mid 11X,$ $X \rightarrow W.$ After simplification convert the grammar into an equivalent grammar in Chomsky normal form (7 Marks)	10	CO 3	BL3



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SLOT: F2+TF2

Step1:- Remove λ production.

$$V \rightarrow \lambda$$

Replace V in all other rules with v

$$U \rightarrow ov / owow / o$$

$$V \rightarrow ov / o$$

$$W \rightarrow iw / iix$$

$$X \rightarrow w$$

Removal of unit production

$$X \rightarrow w$$

Replace $X \rightarrow w$ with rules ($w \rightarrow iw / iix$)

$$X \rightarrow iw / iix$$

After removing unit production

$$U \rightarrow o / ov / owow$$

$$V \rightarrow ov / o$$

$$W \rightarrow iw / iix$$

$$X \rightarrow iw / iix$$



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SLOT: F2+TF2

Steps: Removal of useless production.
No useless production in
the above production.

Convert to CNF

$A \rightarrow BC.$

$A \rightarrow a.$

Steps:

1. Introduce new non-terminal for
terminal $a, l.$

Let $T_0 \rightarrow a$

$T_1 \rightarrow l$

2. Replace terminals in rules that
have more than one symbol

$U \rightarrow a|ov|oww$ becomes.

$U \rightarrow T_0|T_0v|T_0ww$

$V \rightarrow T_0v|T_0$

$w \rightarrow T_1T_w|T_1T_1x$

$x \rightarrow T_1w|T_1T_1x.$

Breakdown the long production



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where $Z_1 \rightarrow T_1 T_1$ $U \rightarrow T_0 T_0 V T_0 Y_1$ $Y_1 \rightarrow W W$ $V \rightarrow T_0 V T_0$ $W \rightarrow T_1 W Z_1 X$ $X \rightarrow T_1 W Z_1 X$ $Z_1 \rightarrow T_1 T_1$ $T_0 \rightarrow 0$ $T_1 \rightarrow 1$			
b) Show that the following grammar is ambiguous. What happens for the following string aaaab. Discuss in terms of parse tree. $S \rightarrow AB aaaB,$ $A \rightarrow a Aa,$ $B \rightarrow b.$ (3 Marks)			

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WINTER SEMESTER 2024-2025

SLOT: F2+TF2

3)b) $S \rightarrow AB aaaB$ $A \rightarrow a Aa$ $B \rightarrow b$ Let, we take $aaab$. $S \rightarrow aaaB$ $aaab$ $aaab$ ① $S \rightarrow AB$ ② ∴ The given grammar is ambiguous.			
4. a) Convert the following grammar to an equivalent grammar in GNF $S \rightarrow AB$ $A \rightarrow BB a$ $B \rightarrow AB a$ (6 Marks) Answer :	10	CO 3	BL3

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CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F2+TF2

4 a) answer: $S \rightarrow aBB \mid aB \mid aBA'B \mid aA'B$ $A \rightarrow aB \mid a \mid aBA' \mid aA'$ $A' \rightarrow BB \mid BBA'$ $B \rightarrow aBB \mid aB \mid aBA'B \mid aA'B \mid a$	The string "aaaab" is not produced by the given grammar. so it cannot be produced	b) Consider the context-free languages $L_1 = \{a^{2n} b^k c^n \mid n, k \geq 1\}$. $L_2 = \{a^m b^{2p} c^p \mid m, p \geq 1\}$. What is $L_1 \cap L_2$? Is it a context-free language? Justify briefly. (4 Marks) Answer : $L_1 = a^2b^+c, a^4b^+c^2, a^6b^+c^3, \dots$ $L_2 = a^+b^2c, a^+b^4c^2, a^+b^6c^3, \dots$ $L_1 \cap L_2 = \{a^{2n}b^{2n}c^n \mid n \geq 1\}$ No Its not a CFG because $a^{2n}b^{2n}$ can be computed in PDA but c^n is not computed.	5.	Construct a Non-deterministic push down automata (NPDA) that recognizes the language $L = \{ab(ab)^n x b(ab)^n a \mid n \geq 0, x = \{a, b\}\}$. Note that x is a symbol a or b. Describe the idea of your construction of transition rules and provide the	10	CO 4	BL3
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CONTINUOUS ASSESSMENT TEST - II
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SLOT: F2+TF2

transition rules. Show the configurations made by the constructed NPDA for the input string "ababababa". 5) $\delta(q_0, a, z) = (q_1, az)$ $\delta(q_1, b, a) = (q_0, ba), (q_2, ba)$ $\delta(q_0, a, b) = (q_1, ab)$ $\delta(q_2, (a b), \epsilon) = (q_3, b)$ $\uparrow$ $\delta(q_3, b, b) = (q_4, \lambda)$ $\delta(q_4, a, a) = (q_4, \lambda)$ $\delta(q_4, b, b) = (q_4, \lambda)$ $\delta(q_4, \lambda, z) = (q_R, z)$ The string "ababababa" is accepted in <u>NPDA</u>.	
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School of Computer Science and Engineering

Winter Semester 2023-2024

Continuous Assessment Test – II

SLOT: A2+TA2

Programme Name & Branch : B.Tech (BCB/BCE/BCI/BCT/BDS/BKT)

Course Name & code: Design and Analysis of Algorithms, BCSE204L

Class Number (s): ALL

Faculty Name (s): ALL

Exam Duration: 90 Min.

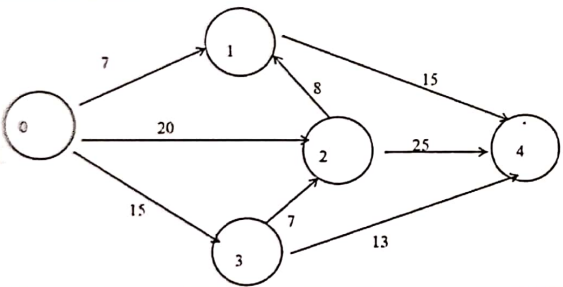
Maximum Marks: 50

Answer All

Q.No.	Question	Max Marks															
1.	A bank has 14 million dollars, which can be invested into stocks of four companies (1, 2, 3, and 4). The following table shows the net revenue of each company and the amount that must be invested into each company. <table><tr><td>Company</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Revenue (Million \$)</td><td>16</td><td>22</td><td>12</td><td>8</td></tr><tr><td>Investment Amount (Million \$)</td><td>5</td><td>7</td><td>4</td><td>3</td></tr></table> The objective for the bank is to select a set of companies for investment, so as to maximize the total revenue with the condition that no partial investment can be done i.e., for each company we can either invest into it or not. Solve the problem with the algorithm whose exponential worst-case time complexity can be improved by employing better techniques for efficient pruning.	Company	1	2	3	4	Revenue (Million \$)	16	22	12	8	Investment Amount (Million \$)	5	7	4	3	10
Company	1	2	3	4													
Revenue (Million \$)	16	22	12	8													
Investment Amount (Million \$)	5	7	4	3													

2.	i) Find out the longest <u>proper prefix</u> that is also the <u>proper suffix</u> for the pattern “ababa”. ii) Calculate the LPS table of the pattern “ababaa”. Using the table constructed in (ii), answer the following. iii) What is the LPS value for the 4th character ‘b’? What does that value infer? iv) Let ‘i’ and ‘j’ be used to denote the index of the characters to be compared in text and pattern respectively. For the text “ababaabababbb”, if the comparison starts from the beginning of the text and if all the characters of the pattern matched with those of the text, what will be the new values of <i>i</i> and <i>j</i>, to proceed with the next comparison? v) For the text “abababababbb”, if all the characters of the pattern except the last, matched with those of the text, what will be the new values of <i>i</i> and <i>j</i>, to proceed with the next comparison?	10
3.	A) <pre> RELAX(<i>u</i>, <i>v</i>, <i>w</i>) 1 if <i>v</i>.<i>d</i> > <i>u</i>.<i>d</i> + <i>w</i>(<i>u</i>, <i>v</i>) 2 <i>v</i>.<i>d</i> = <i>u</i>.<i>d</i> + <i>w</i>(<i>u</i>, <i>v</i>) BELLMAN-FORD(<i>G</i>, <i>w</i>, <i>s</i>) 1 INITIALIZE-SINGLE-SOURCE(<i>G</i>, <i>s</i>) 2 for <i>i</i> = 1 to <i>G</i>.<i>V</i> - 1 3 for each edge (<i>u</i>, <i>v</i>) ∈ <i>G</i>.<i>E</i> 4 RELAX(<i>u</i>, <i>v</i>, <i>w</i>) 5 for each edge (<i>u</i>, <i>v</i>) ∈ <i>G</i>.<i>E</i> 6 if <i>v</i>.<i>d</i> > <i>u</i>.<i>d</i> + <i>w</i>(<i>u</i>, <i>v</i>) 7 return FALSE 8 return TRUE </pre> In the above Bellman-Ford algorithm, make the required modifications to merge the two loops in line numbers 2 and 5 into a single loop and to implement early termination of the loop, if no edges are relaxed in a particular iteration. B) The capacity matrix for a flow network is given below. The first augmenting path found using Edmonds-Karp algorithm for the graph is 0-1-2-4-5. Is this correct or not? Justify your answer.	5

	0	1	2	3	4	5	5
0	0	16	13	0	0	0	
1	0	0	0	12	0	0	
2	0	0	0	0	14	0	
3	0	0	9	0	0	20	
4	0	0	0	7	0	4	
5	0	0	0	0	0	0	

4.	Apply Push-Relabel algorithm for the flow network given below to find the maximum flow that can pass from the source to sink. 	10
5.	Two line-segments intersect if and only if either (or both) of the following conditions holds: 1. Each segment straddles the line containing the other. 2. An endpoint of one segment lies on the other segment. (This condition comes from the boundary case.) <pre> SEGMENTS-INTERSECT(p_1, p_2, p_3, p_4) 1 $d_1 \leftarrow \text{DIRECTION}(p_3, p_4, p_1)$ 2 $d_2 \leftarrow \text{DIRECTION}(p_3, p_4, p_2)$ 3 $d_3 \leftarrow \text{DIRECTION}(p_1, p_2, p_3)$ 4 $d_4 \leftarrow \text{DIRECTION}(p_1, p_2, p_4)$ 5 if $((d_1 > 0 \text{ and } d_2 < 0) \text{ or } (d_1 < 0 \text{ and } d_2 > 0)) \text{ and}$ $((d_3 > 0 \text{ and } d_4 < 0) \text{ or } (d_3 < 0 \text{ and } d_4 > 0))$ 6 return TRUE 7 elseif $d_1 == 0$ and ON-SEGMENT(p_3, p_4, p_1) 8 return TRUE 9 elseif $d_2 == 0$ and ON-SEGMENT(p_3, p_4, p_2) 10 return TRUE 11 elseif $d_3 == 0$ and ON-SEGMENT(p_1, p_2, p_3) 12 return TRUE 13 elseif $d_4 == 0$ and ON-SEGMENT(p_1, p_2, p_4) 14 return TRUE 15 else return FALSE DIRECTION(p_i, p_j, p_k) 1 return $(p_k - p_i) \times (p_j - p_i)$ ON-SEGMENT(p_i, p_j, p_k) 1 if $\min(x_i, x_j) \leq x_k \leq \max(x_i, x_j)$ and $\min(y_i, y_j) \leq y_k \leq \max(y_i, y_j)$ 2 return TRUE 3 else return FALSE </pre>	

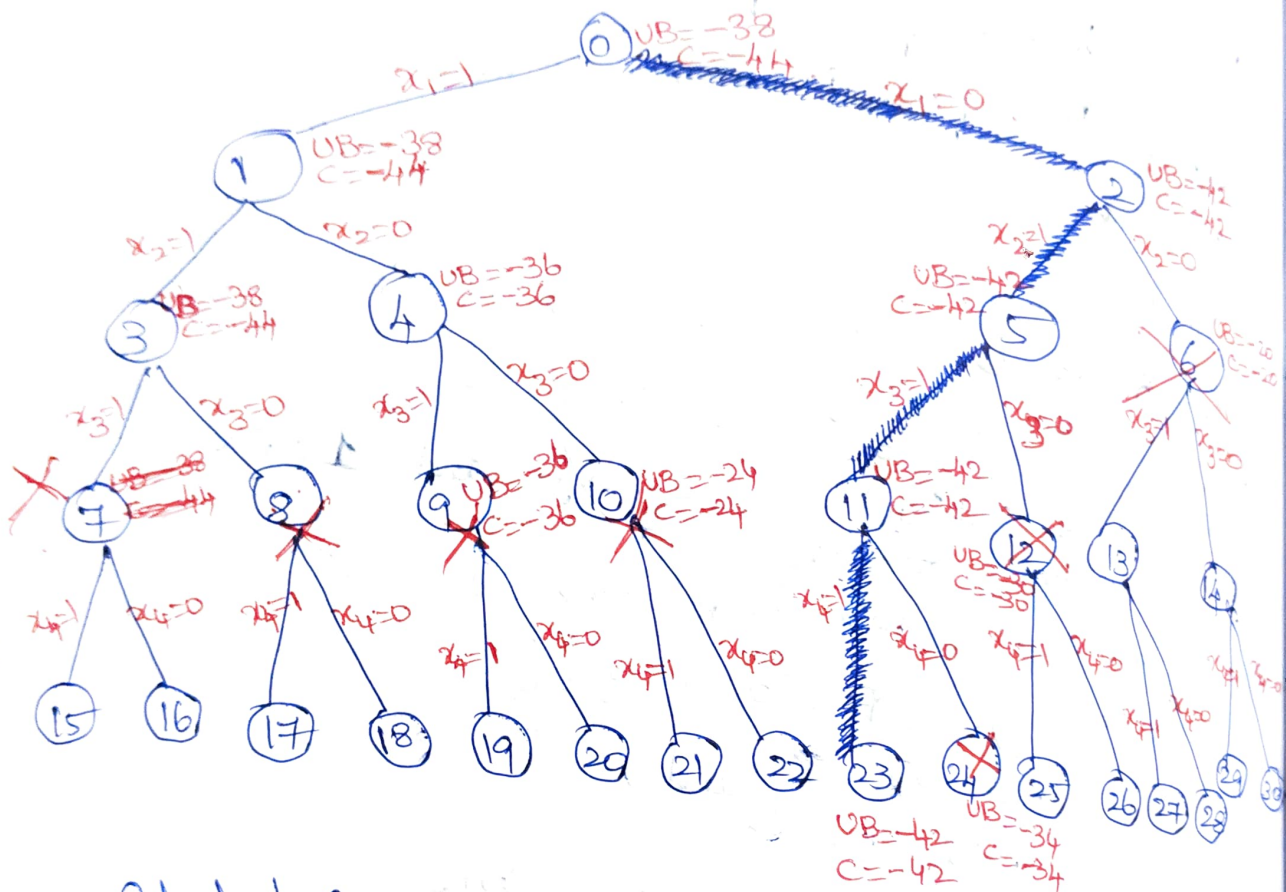
	Given a line-segment L1 with two end-points (2,1) and (6,5), select the other appropriate line-segment among the following to justify your answer for the statements (A), (B) and (C) given below. L2 with two end-points (3,5) and (5,2) L3 with two end-points (3,5) and (4,4) L4 with two end-points (7,6) and (8,7) L5 with two end-points (7,6) and (5,8) L6 with two end-points (5,4) and (7,3)	
	Statement A: The condition in line number 5 of the algorithm is “if $((d_1 > 0 \text{ and } d_2 < 0) \text{ or } (d_1 < 0 \text{ and } d_2 > 0))$ <u>and</u> $((d_3 > 0 \text{ and } d_4 < 0) \text{ or } (d_3 < 0 \text{ and } d_4 > 0))$ return TRUE” Can the “and” operator which is underlined and highlighted in bold, be replaced with “or” operator to prove that one segment straddles the line containing the other?	3
	Statement B: To prove intersection, is it a must to check the condition “ON_SEGMENT(P_i, P_j, P_k)” in line numbers 7 to 14 of the algorithm?	3
	Statement C: The else-if conditions in line numbers 7 to 14 of the algorithm show that any one of the directions d_1, d_2, d_3, d_4 can be zero to prove intersection. Is there any line-segment in the list for which only one direction is zero and does not intersect with L1? Is there any line-segment in the list for which more than one direction is zero and still does not intersect with L1?	4

A2 - CAT-II - Key

1) B & B:

$$UB = -38$$

$$-42$$



Selected Companies: 2, 3, 4.

Total revenue: $22 + 12 + 8 = 42$ million \$

2) i) ababa

Proper prefix

a
ab

aba

abab.

Proper suffix

a

ba

aba

baba

The longest proper prefix that is also the proper suffix is "aba"

(ii) LPS table for "ababaa"

0	1	2	3	4	5
a	b	a	b	a	a
0	0	1	2	3	0

(iii) LPS value of "b" = 2.

It means that the longest proper prefix-suffix for the substring "abab" is "ab" whose length is stored in this cell.

(iv)

	0	1	2	3	4	5	6	7	8	9	10	11	12
Text	a	b	a	b	a	a	b	a	b	a	a	b	b
Pattern	a	b	a	b	a	a							

If array indexing starts from '0' after all the characters of the pattern match with the text, $i=6$ and $j=0$ to start comparing the encircled letter 'b' in text and 'a' in pattern.

(v)

	0	1	2	3	4	5	6	7	8	9	10	11
text	a	b	a	b	a	b	a	b	a	a	b	b
pattern	a	b	a	b	a	a						

5th characters do not match.

$$\therefore j = \text{LPS}[5-1] \\ = \text{LPS}[4] = 3$$

$$\text{So, } i=5, j=3$$

	0	1	2	3	4	5
text	a	b	a	b	a	b
pattern	a	b	a	b	a	a

Characters 'b' in text & pattern will be compared.

3) a)

boolean RELAX(u, v, w)

if $v.d > u.d + w(u, v)$

$v.d = u.d + w(u, v)$

return true

return false

BELLMAN-FORD(G, w, s)

INITIALIZE-SINGLE-SOURCE(G, s)

boolean cycle = false

for $i = 1$ to $|G.V|$

Count = 0

for each edge $(u, v) \in G, E$

if **RELAX(u, v, w)**

if $(i == v)$

cycle = true

break

else

Count = Count + 1

if (Count == 0)

return true

if (cycle == true)

return false

else

return true

b) Edmonds-Karp:

First augmenting path:

Queue:

~~0~~ ~~1~~ ~~3~~ ~~2~~ ~~4~~ $\emptyset$

Node Id	0	1	2	3	4	5
Visit state	1	1	1	1	1	1
Predecessor	-1	0	0	1	2	3

Aug. path = $5 \leftarrow 3 \leftarrow 1 \leftarrow 0$

$\Rightarrow 0 \rightarrow 1 \rightarrow 3 \rightarrow 5$

So, $0-1-2-4-5$ is not the correct augmenting path.

A) Push-Relabel:

Node Id	0	1	2	3	4
Height	5	0 1	0 1	0 2	0
Excess Flow	-	7 0	20 0	15 0	0 42

Capacity

	0	1	2	3	4
0	0	7	20	15	0
1	0	0	0	0	15
2	0	8	0	0	25
3	0	0	7	0	13
4	0	0	0	0	0

Flow

	0	1	2	3	4
0	0	7 0	20 0	15 0	0
1	-7 0	0	0	0	0 7
2	-20 0	0	0	-2 0	22 20
3	-15 0	0	2 0	0	0 13
4	0	-7 0	-20 0	-13 0	0

After preflow:

i) Excess flow node = 1; ~~Excess flow node = 1~~

Relabel node 1 $\Rightarrow Ht(1) = 0 + 1 = 1$

Flow from 1 to 4 = $\min(7, 15) = 7$ ✓

ii) Excess flow node = 2

Relabel node 2 $\Rightarrow Ht(2) = 0 + 1 = 1$

Flow from 2 to 4 = $\min(20, 25) = 20$ ✓

iii) Excess flow node = 3

Relabel node 3 $\Rightarrow Ht(3) = 0 + 1 = 1$

Flow from 3 to 4 = $\min(15, 13) = 13$ ✓

iv) Excess flow node = 3

Relabel node 3 $\Rightarrow Ht(3) = 1 + 1 = 2$

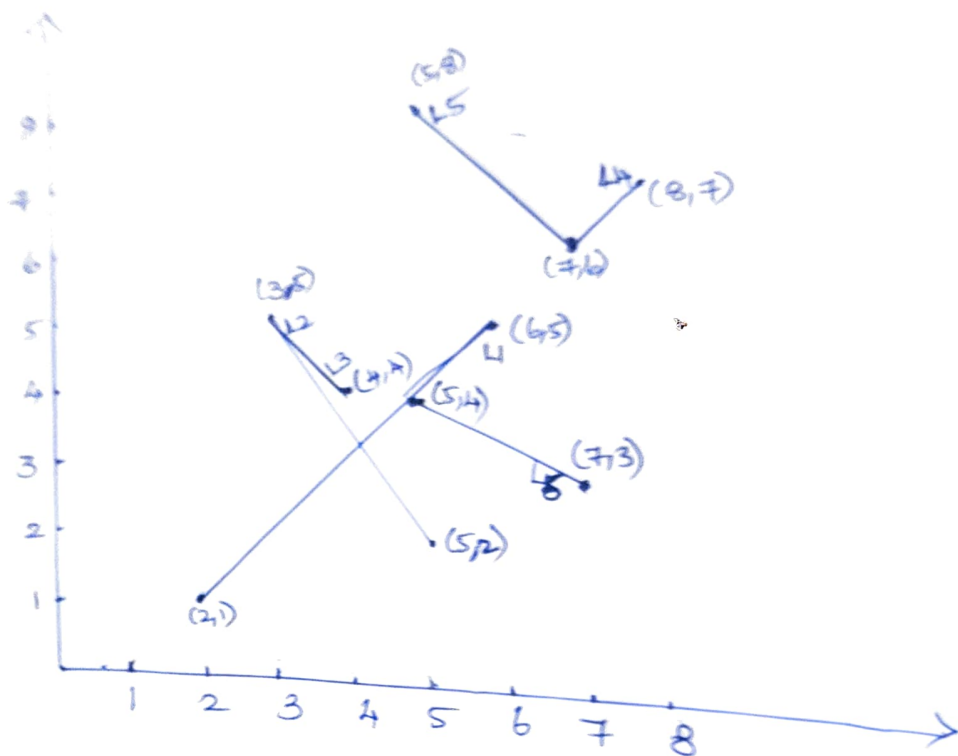
Flow from 3 to 2 = $\min(7, 2) = 2$

v) Excess flow node = 2

Flow from 2 to 4 = $\min(2, 25 - 20) = 2$ ✓

No more excess flow nodes. Maxflow = $7 + 20 + 13 + 2 = 42$

11. (2,1) & (6,5)



Line segment $d_1 = \text{dis}(P_3, P_4, P_1)$ $d_2 = \text{dis}(P_3, P_4, P_2)$ $d_3 = \text{dis}(P_1, P_2, P_3)$ $d_4 = \text{dis}(P_1, P_2, P_4)$

L_2
 $(3,5), (5,2)$ $\begin{vmatrix} -1 & -2 \\ -4 & -3 \end{vmatrix} = 11$ $\begin{vmatrix} 3 & 2 \\ 0 & -3 \end{vmatrix} = -9$ $d_3 = \begin{vmatrix} 1 & 4 \\ 4 & 4 \end{vmatrix} = -12$ $\begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix} = 8$

L_3
 $(3,5), (4,4)$ $\begin{vmatrix} -1 & 1 \\ -4 & -1 \end{vmatrix} = 5$ $\begin{vmatrix} 3 & 1 \\ 0 & -1 \end{vmatrix} = -3$ $d_3 = \begin{vmatrix} 1 & 4 \\ 4 & 4 \end{vmatrix} = -12$ $\begin{vmatrix} 2 & 4 \\ 3 & 4 \end{vmatrix} = -4$

L_4
 $(7,6), (8,7)$ $\begin{vmatrix} -5 & 1 \\ -5 & 1 \end{vmatrix} = 0$ $\begin{vmatrix} -1 & 1 \\ -1 & 1 \end{vmatrix} = 0$ $\begin{vmatrix} 5 & 4 \\ 5 & 4 \end{vmatrix} = 0$ $\begin{vmatrix} 6 & 4 \\ 6 & 4 \end{vmatrix} = 0$

L_5
 $(7,6), (5,8)$ $\begin{vmatrix} -5 & -2 \\ -5 & 2 \end{vmatrix} = -20$ $\begin{vmatrix} -1 & -2 \\ -1 & 2 \end{vmatrix} = 4$ $\begin{vmatrix} 5 & 4 \\ 5 & 4 \end{vmatrix} = 0$ $\begin{vmatrix} 3 & 4 \\ 7 & 4 \end{vmatrix} = -16$

L_6
 $(5,4), (7,3)$ $\begin{vmatrix} -3 & 2 \\ -3 & -1 \end{vmatrix} = 9$ $\begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = -3$ $\begin{vmatrix} 3 & 4 \\ 3 & 4 \end{vmatrix} = 0$ $\begin{vmatrix} 5 & 4 \\ 2 & 4 \end{vmatrix} = 12$

Statement A:

If "and" is replaced by "or" according to the crossproduct, for L3, $d_1 > 0$ and $d_2 < 0$ but d_3 and d_4 are both negative; But still because of "or" L3 will be reported as intersecting with L1. From figure, it can be seen that this result contradicts.

Statement B:

For L5, $d_3 = 0$. Let's check onsegment (P_1, P_2, P_3) .

onsegment (P_1, P_2, P_3) : $P_3 = (7, 6)$, $P_1 = (2, 1)$, $P_2 = (6, 5)$

onsegment $\Rightarrow$ ~~$2 < 7$~~ and ~~$2 < 7 \times 6$~~ and ~~$1 < 6 \times 5$~~ $\Rightarrow$ false

For L6, $d_3 = 0$. Let's check onsegment (P_1, P_2, P_3)

$P_3 = (5, 4)$, $P_1 = (2, 1)$, $P_2 = (6, 5)$

$2 < 5 < 6 \checkmark$ and

$1 < 4 < 5 \checkmark$

$\therefore$ onsegment $(P_1, P_2, P_3) = \text{true}$.

From figure, it can be seen that L5 does not intersect with L1 but L6 intersects, although in both $d_3 = 0$. This proves that "onsegment" condition is a must.

Statement C:

For L5, $d_3 = 0$ but still does not intersect with L1 as onsegment (P_1, P_2, P_3) is false.

For L4, all direction values are zeroes.

onsegment (P_3, P_4, P_1) : ~~$7 \times 2 < 8$~~ $\Rightarrow$ false

onsegment (P_3, P_4, P_2) : ~~$7 \times 6 < 8$~~ $\Rightarrow$ false

onsegment (P_1, P_2, P_3) : ~~$2 < 7 \times 6$~~ $\Rightarrow$ false

onsegment (P_1, P_2, P_4) : ~~$2 < 8 \times 6$~~ $\Rightarrow$ false

$\therefore$ L4 also does not intersect with L1

$\longrightarrow 0$



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F2+TF2

Programme Name & Branch : B.Tech CSE and Specialization
Course Code and Course Name : BCSE304L Theory of Computation
Faculty Name(s) : Prof. Kannadasan R, Prof. Mohana C M, Prof. Annapurna
Jonnalagadda, Prof. Navamani T M, Prof. Arpan Garai, Prof. Anand Bihari, Prof. Uma Priya D
, Prof. Saravanan R, Prof. Naga Priyadarsini R, Prof. Krishnaraj N, Prof. Gunavathi C, Prof.
Viswanathan P, Prof. Vijayarajan V, Prof. Vishnu Priya A, Prof. Sridevi S, Prof. Prabakaran N,
Prof. Sri Preethaa K R, Prof. Katari Balakrishna
Class Number(s) : VL2024250501656, 1634, 1622, 1628, 1648, 1632,
1644, 1640, 1654, 1642, 1618, 1620, 1616, 1638, 1636, 1624, 1650, 1614
Date of Examination : 21.03.2025 FN
Exam Duration : 90 minutes
Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcomes(CO):
CO2: Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO3: Identify limitations of some computational models and possible methods of proving them.
CO4: Represent the abstract concepts mathematically with notations.

Q. No	Question	M	CO	BL
1.	(a) Convert the following Grammar into Finite automata $S \rightarrow aA bB$ $A \rightarrow aA bS c$ $B \rightarrow bB a$ (3 Marks) (b) Let R_1 and R_2 be regular languages over Σ . Then, exclusive of R_1 and R_2 is defined as $EXC(R_1, R_2) = \{ x \in \Sigma^* : x \text{ in } \{R_1 - R_2\} \text{ or } x \text{ in } \{R_2 - R_1\} \}$. If $x \text{ in } \{R_i - R_j\}$ means x is in R_i and not in R_j . Prove that $EXC(R_1, R_2)$ is regular. (3 Marks) (c) Using pumping lemma, demonstrate that the given language $L = \{x^i y^j z^k \mid k \leq i+j\}$ is not regular. (4 Marks)	10	CO 2	BL2
2.	Consider the following context free grammar (where S is the start symbol) $S \rightarrow AB BC$ $A \rightarrow BA a$ $B \rightarrow CC b$ $C \rightarrow AB c$ Apply CYK algorithm with all required steps for the input string "abbca".	10	CO 3	BL3
3.	a) Consider the following grammar (where U is the start symbol) remove all unit-productions, all useless productions, and all λ - productions from the grammar where $\Sigma = \{0,1\}$ $U \rightarrow 0V 0WW,$ $V \rightarrow 0V \lambda,$ $W \rightarrow 1W 11X,$ $X \rightarrow W.$	10	CO 3	BL3



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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F2+TF2

	After simplification convert the grammar into an equivalent grammar in Chomsky normal form (7 Marks)			
	b) Show that the following grammar is ambiguous. What happens for the following string aaaab. Discuss in terms of parse tree. $S \rightarrow AB aaaB,$ $A \rightarrow a Aa,$ $B \rightarrow b.$ (3 Marks)			
4.	a) Convert the following grammar to an equivalent grammar in GNF $S \rightarrow AB$ $A \rightarrow BB a$ $B \rightarrow AB a$ (6 Marks)	10	CO 3	BL3
	b) Consider the context-free languages $L1 = \{a^{2n} b^k c^n \mid n, k \geq 1\}$. $L2 = \{a^m b^{2p} c^p \mid m, p \geq 1\}$. What is $L1 \cap L2$? Is it a context-free language? Justify briefly. (4 Marks)			
5.	Construct a Non-deterministic push down automata (NPDA) that recognizes the language $L = \{ab(ab)^n x b(ab)^n a \mid n \geq 0, x = \{a, b\}\}$. Note that x is a symbol a or b . Describe the idea of your construction of transition rules and provide the transition rules. Show the configurations made by the constructed NPDA for the input string "ababababa".	10	CO 4	BL3

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SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS**CAT-2****WINTER SEMESTER 2025-2026****SLOT:B2+TB2**

Programme Name & Branch : B.Tech CSE (AIDE and CS)
Course Code and Course Name : BCSE304L Theory of Computation
Faculty Name(s) : Dr. Uma Maheswari G, Dr. Raghavan R
Class Number(s) : VL2025260503589, VL2025260503591
Date of Examination : 16/3/2026
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcomes
 2. Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata
 3. Identify limitations of some computational models and possible methods of proving them
 4. Represent the abstract concepts mathematically with notations

Q. No	Question	M	CO	BL
1.	Give a regular expression for each of the following languages defined on the $\Sigma = \{0,1\}$ a) $L = \{w w \text{ contains the substring } 001\}$ b) $L = \{w w \text{ end with '101' and not containing the substring '111'}\}$ c) $L = \{w w \text{ contain odd number of 0's}\}$ d) $L = \{w w \text{ contain at least three 1's}\}$ e) $L = \{w w \text{ contain at least one 0 and at least one 1}\}$	10	2	3
2.	Apply pumping lemma to prove $L = \{a^ib^jc^k \mid k \geq i+j\}$ is not regular.	10	2	3
3.	a) For the given regular expression, construct an NFA with empty transition recognizing the language $(0+1)(01)^*(011)^*$.	5	2	3
	b) Which of the following is false with proof? i) $(a^*b)^*a^* = a^*(ba^*)^*$ ii) $(aa)^*(\epsilon + a) = a^*a$	5		
4.	a) Find Context Free Grammar G for the following language $L = \{a^n b^m c^m d^n \mid n, m \geq 1\}$	5	3	3
	b) Check the following grammar is ambiguous or not.	5		



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REG.NO.:

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

CAT-2
WINTER SEMESTER 2025-2026

SLOT:B2+TB2

	$G = (V, T, P, S), V = \{S, X\}, T = \{a, b\},$ $P = \{S \rightarrow XX, X \rightarrow XXX \mid a \mid bX \mid Xb\}$			
5	a) Convert the following grammar to Chomsky Normal Form $S \rightarrow \epsilon \mid (S) \mid SS$	3		
	b) Obtain Griebach Normal Form equivalent for the Context Free Grammar given below. $S \rightarrow ABA$ $A \rightarrow aA \mid \epsilon$ $B \rightarrow bB \mid \epsilon$	7	4	3



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

Programme Name & Branch : B.Tech, CSE
Course Code and Course Name : BCSE304L – Theory of Computation
Faculty Name(s) : KANAGARAJ R, MADIAJAGAN M, SATHIYA KUMAR C, KARTHIK G M, BASKARAN P, PARTHASARATHY G, SARITHA MURALI, RAJARAJAN G, SHALINI L, UMA PRIYA D, BOOMINATHAN P, LAKSHMANAN K, BHAWANA TYAGI, BHUVANESWARI M, IYAPPAN P, ISLABUDEEN M, PRAKASH M, SATHYA K, ADRIJA BHATTACHARYA, DEBI PRASANNA ACHARIYA, K.KRISHNA RANI SAMAL, SUGANTHINI C
Class Number(s) : VL2024250501633, 1615, 1619, 1625, 1637, 1639, 1635, 1653, 1631, 1643, 1617, 1651, 1649, 1645, 1627, 1647, 1629, 1641, 1623, 1613, 1621, 1655
Date of Examination : 21-Mar-2025, 2:00 PM – 3.30 PM
Exam Duration : 90 minutes Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Q. No	Question	M	CO	BL
1.	a) Prove that the language $L = \{a^n b^m \mid 1 \leq n < m\}$ is regular language or not. b) Let h be a function defined as $h: \Sigma \rightarrow \Gamma^*$ (i.e., letter to word substitution) and is called homomorphism. The function h is extended for words with $h(ab) = h(a).h(b)$ for $a, b \in \Sigma$. The inverse homomorphism function h^{-1} is defined in a usual way. Consider the following homomorphism h from the alphabet $\{0, 1, 2\}$ to $\{a, b\}$ defined by: $h(0) = ab, h(1) = b, h(2) = aa$. 1. What is $h(0210)$? 1 mark 2. If L is the language whose regular expression is $(ab + baa) \cdot bab$, what is $h^{-1}(L)$? 2 marks	7		
2.	a) Consider the grammar $X \rightarrow aXb \mid XX \mid \epsilon$. Prove that the grammar is ambiguous. b) Design context free grammar for the language $L = \{a^i b^j c^k \mid i = j + k\}$ and derive the string $a^4 b^3 c$ using your grammar. c) Consider the grammar G with $S \rightarrow 0S1 \mid 1S0 \mid \epsilon$. In this grammar, for every 0 there is a 1 and vice versa. Is this grammar generating $L = \{w \mid n_0(w) = n_1(w)\}$? Justify. If not, what further modifications are required in the grammar in order to generate the language L ?	3 4 3	2 3	3 2
3.	a) Consider the grammar $S \rightarrow ASA \mid aB, A \rightarrow B, B \rightarrow b \mid \epsilon$. Convert into Chomsky Normal Form b) Consider the context-free grammar whose rules are $S \rightarrow SS \mid Sa \mid a$. Convert into equivalent grammar in GNF.	6 4	3	4
4.	Given the context-free grammar $G: S \rightarrow XY, X \rightarrow YY \mid 0, Y \rightarrow XY \mid 1$. Use the CYK algorithm to determine whether the strings 00110 is in the language generated by the given grammar.	10	3	3
5.	a) Consider the PDA for accepting palindrome strings over $\{a, b\}$. For the below incomplete trace sequence (instantaneous description – ID) and moves for the given string $ababababa$, identify the required state or partial input string or	5	1	4



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CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

stack content in the "fill in the blanks" entry. Note q_0 is the start state and q_f is the final state, Z_0 is the start stack symbol.

- $(q_0, ababababa, Z_0)$
- $\vdash (q_0, babababa, aZ_0)$
- $\vdash (q_0, abababa, baZ_0)$
- $\vdash (q_0, bababa, -)$
- $\vdash (q_0, ababa, -)$
- $\vdash (-, baba, babaZ_0)$
- $\vdash (q_1, -, abaZ_0)$
- $\vdash (q_1, ba, baZ_0)$
- $\vdash (q_1, a, aZ_0)$
- $\vdash (-, \epsilon, Z_0)$
- $\vdash (q_f, \epsilon, Z_0)$

b) Consider the language $L = \{a^{2n}c^mb^n \mid n, m \geq 1\}$. The objective is to construct a (pushdown automata) PDA for this language and below are the partial transition rules. The main idea is to push a 0 in the stack whenever a is read and when b s are read, they need to be cancelled against 0s such a way that for every one b is read, two 0s need to be cancelled out in the stack. You are expected to fill-in the remaining rules which are left un-attempted (denoted with - in the rules) and complete the rules such that the constructed PDA will accept the given language L . FYI, the transition function δ of the PDA is defined as: $\delta : Q \times \Sigma \cup \{\lambda\} \times \Gamma \rightarrow \text{finite subset of } Q \times \Gamma^*$. You are allowed to use only one (new) additional state which is q_f , the final state. Write only incomplete rules 2,4,6,7,8 for answering the question. Z is the start stack symbol.

1. $\delta(q_0, a, Z) = (q_1, 0Z)$
2. $\delta(q_1, a, -) = (q_1, 00)$
3. $\delta(q_1, c, 0) = (q_2, 0)$
4. $\delta(q_2, c, -) = (-, -)$
5. $\delta(q_2, b, 0) = (q_3, \lambda)$
6. $\delta(q_3, -, -) = (-, \lambda)$
7. $\delta(q_4, b, -) = (-, -)$
8. $\delta(q_4, \lambda, -) = (-, -)$

5

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2025-2026

SLOT: B1+TB1

Programme Name & Branch	: B.Tech & CSE (AIDE and CS)	
Course Code and Course Name	: BCSE304L and Theory of Computation	
Faculty Name(s)	: Dr. Uma Maheswari G, Dr Pradeepa^M, Dr. Raghavan R	
Class Number(s)	: VL2025260503588, VL2025260503565, VL2025260503590	
Date of Examination	: 16.3.2026	
Exam Duration	: 90 minutes	Maximum Marks: 50

General instruction(s):

Answer All Questions

Include steps (mandatory) and Diagrams (if needed)

M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Course Outcomes

2. Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata
3. Identify limitations of some computational models and possible methods of proving them
4. Represent the abstract concepts mathematically with notations

Q. No	Question	M	CO	BL
1a	$L = \{ w \in \{a, b\}^* \mid w \text{ contains } aa \text{ as a substring} \}$. Find complement of L and Prove that regular languages are closed under complementation			
1b	Let $\Sigma = \{ a, b, \dots, z, A, B, \dots, Z \}$ and consider the languages over Σ as Noun, Verb and The as mentioned below Noun = { Puppy, Rainbow, Whale, ... } Verb = { Hugs, Juggles, Loves, ... } and The = { The } Prove that the regular languages are closed under concatenation.	5 5	CO2	3
2a	Check whether the given language is ambiguous or inherently ambiguous	5		
2b	$L = \{ 0^i 1^j 2^k \mid i=j \text{ or } j=k \}$ Show that $L = \{ 0^p \mid p \text{ is prime} \}$ is not regular using pumping lemma. Include all the associated steps. (PTO)	5	CO2	2



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(Deemed to be University under section 3 of U.G. Act, 1956)

REG.NO.:

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2025-2026

SLOT: B1+TB1

3a	Convert the following Deterministic Finite Automaton to Regular Grammar (q_0 is the initial state and q_4, q_5, q_6 are final states).	5			
3b	Convert the obtained regular grammar in question 3a to the corresponding regular expression.	5			
4	Bring the grammar G with $V = \{S, A, B\}$, $T = \{a, b\}$ and productions P $S \rightarrow A$ $A \rightarrow a B a \mid a$ $B \rightarrow b A b \mid b$ into Greibach Normal Form.	10	CO4	3	
5a	Find the regular expression for the following DFA using Arden's theorem.	5	CO2	3	
5b	Construct a Context Free Grammar for $(0+1)^*111$. Verify the constructed grammar using valid strings.	5			

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Final Assessment Test – April 2025
Course: BCSE204L - Design and Analysis of Algorithms
Class NBR(s): 1459/1472/1489/1496/1504/1510/1518/
1525/1530/1534/1537/1543/1547/1555/
1561/1567/1578/1586/1594/1600/1608

Reg. No: 23BCE0990

Slot: A1+TA1

Max. Marks: 100

Time: Three Hours

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions
(10 X 10 = 100 Marks)

1. The Karatsuba algorithm is a divide-and-conquer approach for fast integer multiplication. Break down the Karatsuba algorithm into its constituent stages: problem decomposition, recursive subproblems, and recombination. Derive and analyze its time complexity, comparing it with the standard multiplication method. Illustrate the algorithm for the multiplication of two integers 21 and 143.

2. Construct the optimal Huffman code for a given set of character frequencies derived from the first 8 prime numbers.

Character	a	b	c	d	e	f	g	h
frequency	2	3	5	7	11	13	17	19

Derive the time and space complexity of the Huffman code. Define the prefix free property and the need for prefix free code in Huffman coding. Show that Huffman coding takes less space as compared to fixed size code.

3. A logistics company wants to optimize the delivery route for a salesperson who must visit four cities exactly once before returning to the starting city. The company has recorded the travel distances between cities in the given matrix in Table 2.

As a software engineer, develop an efficient route-planning algorithm using Dynamic Programming (DP).

- Formulate the Traveling Salesman Problem (TSP) using the DP approach and explain its optimal substructure.
- Using the Held-Karp Algorithm (Dynamic Programming), compute the minimum travel cost for the given distance matrix and show step-by-step calculations.

- c) Compare the time and space complexity of the dynamic programming approach with the brute-force approach.

	A	B	C	D
A	0	4	1	9
B	3	0	6	4
C	4	1	0	2
D	6	5	-4	0

Table 2: Distance matrix between cities.

A delivery company wants to optimize the selection of delivery packages to maximize profit while ensuring that the total weight of selected packages does not exceed the weight limit of their delivery van. Each package has a specific weight and value associated with it.

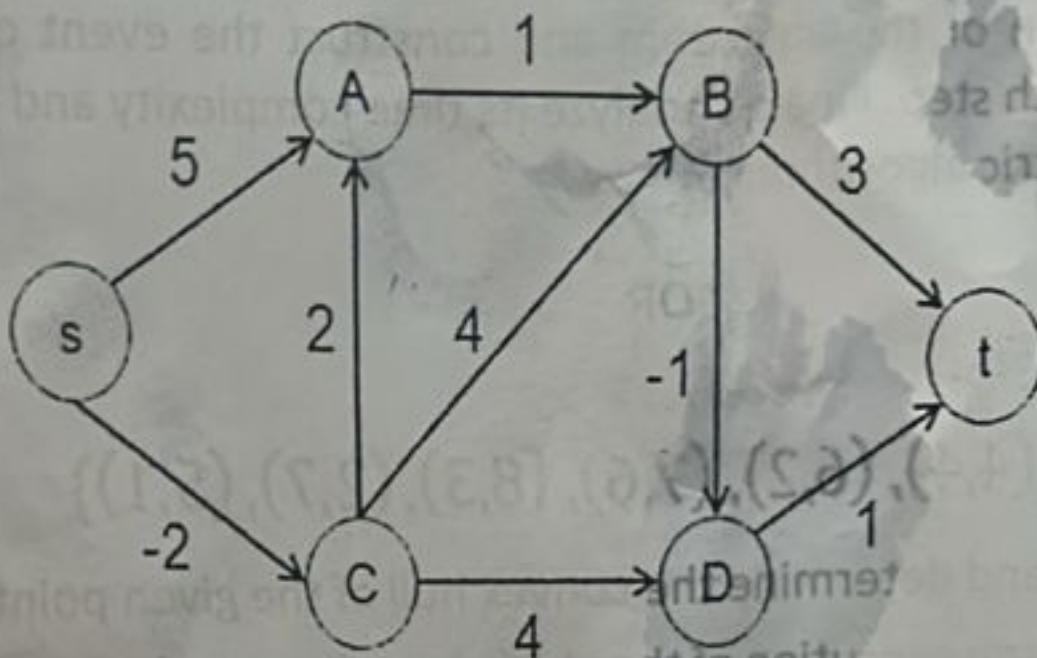
As an algorithm specialist, apply the **FIFO Branch & Bound** approach to solve the **0-1 Knapsack Problem** with the following items:

Package	Weight (kg)	Value (Rs.)
1	2	40
2	3	50
3	4	70
4	5	80

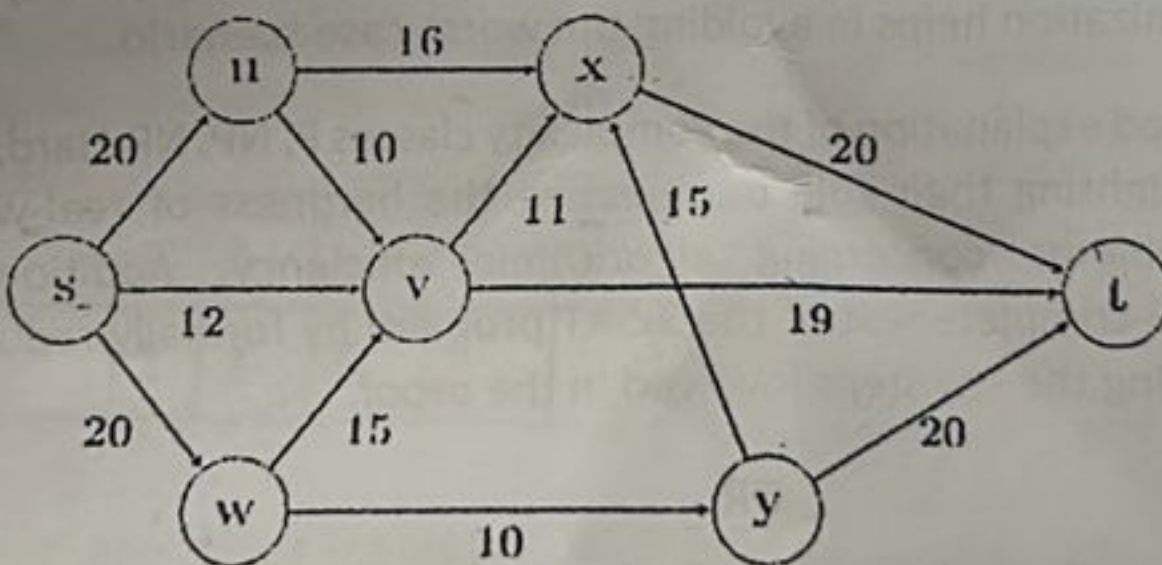
Contrast the Backtracking and Branch & Bound paradigms in solving the 0-1 Knapsack Problem, highlighting their differences in terms of search strategy, pruning mechanism, efficiency, and applicability.

As a bioinformatics researcher, you are tasked with developing an efficient algorithm to identify specific genetic markers in DNA sequences. You need to search for the genetic marker pattern $P = \text{"ACGTACGT"}$ within the target DNA sequence $T = \text{"GCACGTACGTGACGTACGTA"}$. Apply the KMP and Rabin-Karp algorithm for finding all occurrences of the pattern P in the given text T . Trace the execution of your chosen algorithm when searching for pattern P in sequence T , highlighting key comparisons and shifts. For Rabin Karp, consider $A=1, C=2, G=3, T=4$, base 4, and mod 101 and show the number of spurious hits. Compare the efficiency of your solution by calculating the total number of character comparisons performed by both algorithms for this specific example. Identify the algorithm that runs faster as compared to the other.

Apply the Bellman Ford algorithm to compute the all-pairs shortest paths for the given weighted graph. Show the step-by-step execution of the algorithm. Discuss the strength and weakness of the algorithm.



7. A logistics company is planning the most efficient way to transport goods from a central warehouse (source s) to a major distribution center (destination t) through a network of cities: u, v, w, x , and y . Each road connecting these cities has a specific maximum carrying capacity, representing the amount of goods that can be transported at a time, represented by the graph given below.



Currently, no goods are flowing through the network. The company wants to maximize the shipment flow using the **Ford-Fulkerson algorithm** while following these conditions:

- In the first step, the route with the **longest augmenting path** (largest number of cities) should be chosen for transportation.
- In the second step, the route with the **highest residual capacity** should be prioritized.

In your answer, clearly show (i) the original graph with all capacities filled up, (ii) the augmenting path chosen in the first step (write this as a sequence of vertices, do not mark on the graph), (iii) the flow network after the first augmentation, (iv) the corresponding residual graph after the first augmentation, (v) the augmenting path chosen in the second step, (vi) the flow network after the second augmentation, (vii) the corresponding residual graph after the second augmentation. viii) Also write the time and space complexity analysis of the algorithm.

a) Given the set of points $P = \{(1,2), (3,5), (4,4), (6,2), (7,6), (8,3)\}$ and line segments formed by connecting $(1,2) \rightarrow (4,4)$, $(3,5) \rightarrow (7,6)$, $(6,2) \rightarrow (8,3)$, and $(3,5) \rightarrow (6,2)$. Apply the sweep line algorithm to detect intersections among these segments. Demonstrate the step-by-step execution of the algorithm and construct the event queue and status structure at each step. Finally, analyze its time complexity and discuss its significance in geometric algorithms.

OR

b) Given the set of points:

$$P = \{(1,2), (3,5), (4,4), (6,2), (7,6), (8,3), (2,7), (5,1)\}$$

Apply **Graham's Scan** and determine the convex hull of the given points.

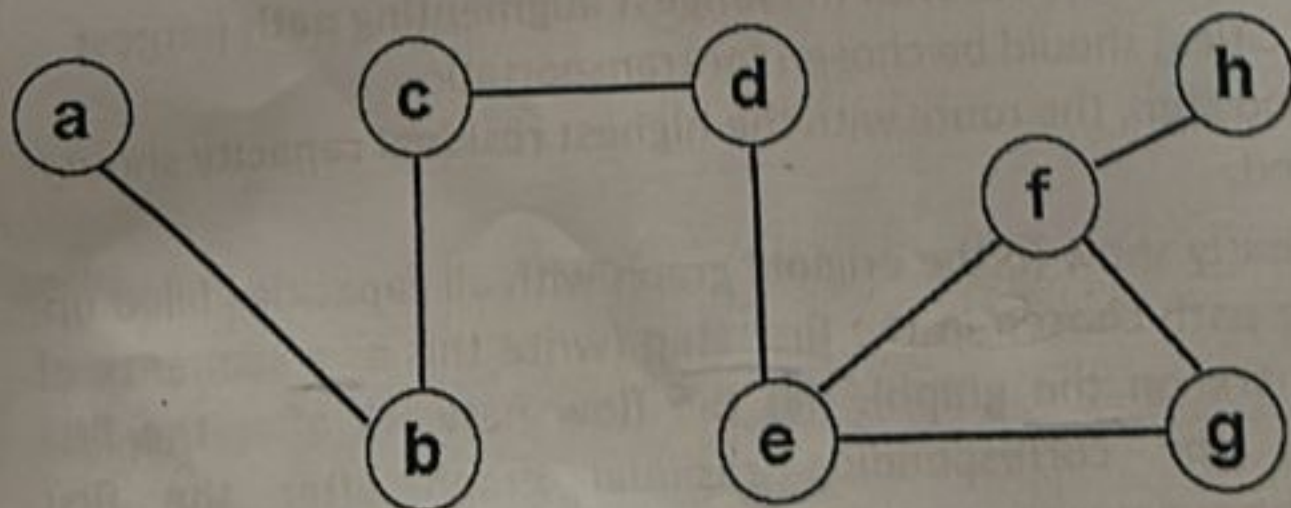
Illustrate the step-by-step execution of the algorithm, showing the intermediate hull formations. Compare computational efficiency of **Graham's Scan** and **Jarvis' March** in terms of **time complexity** and discuss the conditions under which one method may be preferred over the other.

9. Apply the **Randomized QuickSort** algorithm to sort the array $[9, 4, 7, 2, 6, 3, 8]$. Show the step-by-step execution, including the selection of random pivots, partitioning, and recursive calls. Analyze its expected time complexity and discuss how randomization helps in avoiding the worst-case scenario.

10. a) Provide a summarized explanation of the complexity classes P, NP, NP-Hard, and NP-Complete, highlighting their role in assessing the hardness of real-world computational problems concerning algorithmic efficiency. Additionally, demonstrate the NP-completeness of the 3-SAT problem by formally reducing SAT to 3-SAT, outlining the key steps involved in the proof.

OR

10. b) Provide a summarized explanation of the Vertex Cover Problem, its computational complexity, and its significance in real-world applications. Discuss why it is classified as NP-complete and demonstrate the proof that Vertex Cover has a 2-approximation algorithm, outlining the key steps involved in the approximation technique. Obtain a vertex cover for the following graph.



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V/D/TY

Reg. No:



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Final Assessment Test - April 2025

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 1468/1481/1493/1499/1506/1513/1521/
1527/1531/1538/1544/1551/1558/1563/1571/1582/
1589/1597/1606

Slot: A2+TA2

Max. Marks: 100

Time: Three Hours

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1. a) Obtain an upper bound for the following recurrence equation using recursive tree method and verify using Master's theorem [6+4]

$$T(n) = \begin{cases} 1 & \text{when } n = 1 \\ 3T\left(\frac{n}{4}\right) + n^2 & \text{when } n > 1 \end{cases}$$

- b) Solve the following Knapsack problem which consists of divisible items using Greedy strategy. Knapsack capacity is 35. Find the items to be included into the knapsack bag to gain optimal profit, where the weights of items are 10, 20, 12 with corresponding profits 60, 100, 120.

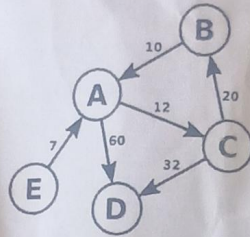
2. Design a divide and conquer algorithm to find the maximum Subarray whose sum is maximum. Consider the array of numbers given by [2, -4, 1, 9, -6, 7, -3] to illustrate the algorithm. Trace the steps of the algorithm with the help of a recursive tree. Discuss its time complexity.

3. Design a dynamic programming algorithm for parenthesization of matrix chain which will require minimum number of scalar multiplications to find the product. Consider the chain of matrices given by A(2x3) B(3x2) C(2x4) and D(4x3) to illustrate the algorithm.

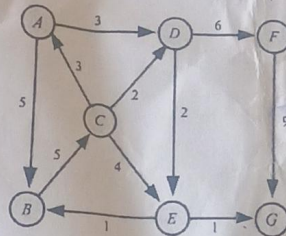
4. Solve the following instance of Knapsack problem using fixed length FIFO branch and bound strategy where we have 4 items and the knapsack bag has a capacity of 15. The profit values are {10, 12, 20, 18} and the corresponding weights are {2, 4, 5, 9}. Draw the annotated state space tree with cost and bound values. The steps for calculating bound values of various nodes should also be listed. Find the optimal solution and provide the list of items to be added into the knapsack bag.

5. Apply an algorithm that constructs longest prefix suffix (LPS) table for the pattern to find where it is plagiarized in a given text. List the algorithm and trace the steps for the pattern given by 'ABABAD' and the text given by 'ABCABABABAD'. Also return the position where the pattern is found in the text. Analyze its time complexity.

6. Find all pairs shortest path for the following weighted digraph using Floyd Warshall's Algorithm. Analyse its time complexity. Illustrate the tracing of the algorithm.



7. Apply Ford Fulkerson's algorithm on the following network flow graph to find the maximum flow from start vertex 'A' to sink vertex 'G'. In each iteration, the graph and the residual flow network obtained after augmenting the flow should be drawn and described.

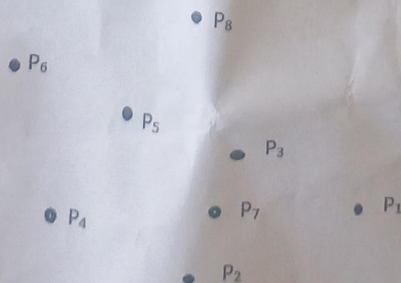


- 8.a) Design an algorithm for checking whether the two given line segments intersect or not. Trace and find if the line segment given by p_1q_1 and another line segment given by p_2q_2 intersect or not.
- $p_1(2,10)$ $q_1(7,2)$ and $p_2(4,15)$ $q_2(13,3)$
 - $p_1(6,4)$ $q_1(10,16)$ and $p_2(8,10)$ $q_2(13,6)$
 - $p_1(6,1)$ $q_1(7,3)$ and $p_2(8,4)$ $q_2(6,7)$
 - $p_1(2,3)$ $q_1(10,3)$ and $p_2(5,2)$ $q_2(7,3)$

OR

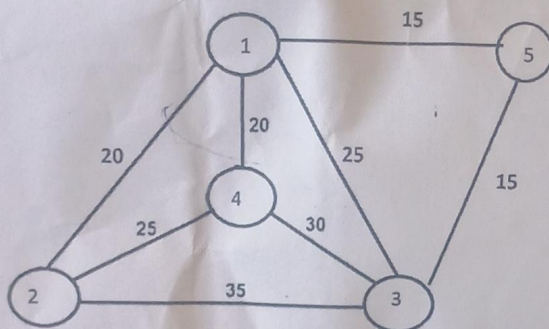
- 8.b) Consider the following set of points 'Q' in a XY plane. Construct a smallest polygon enclosing all the points. Design a $O(n \log n)$ time algorithm to find the same and trace it using the following points. Elaborate the step-by-step process of constructing the polygon.

$$Q = \{P_1(10,3), P_2(6,1), P_3(8,4), P_4(2,2), P_5(3,5), P_6(1,6), P_7(7,3), P_8(6,7)\}.$$



- 9 Apply Randomized quicksort algorithm by selecting the random pivot to trace the steps to sort in detail. Discuss how the average time complexity of randomized quicksort is better than the worst-case time complexity of deterministic quicksort. The elements are given below:
99,34,46,22,60,23,15,14,33,20

- 10.(a) i. Prove that the approximation algorithm for solving Travelling Salesman problem is a polynomial time 2-approximation algorithm. [4]
ii. Design a polynomial time approximation algorithm for the travelling salesman problem. Illustrate the algorithm for the graph given below. Let 1 be the start vertex. [6]



OR

- 10.(b) Prove that "3-CNF (Conjunctive Normal Form) SAT" is NP complete. Use the satisfiability formula given below to prove the NP completeness. Illustrate the three steps for conversion of ϕ to ϕ''' . For two clauses, the detailed steps for converting ϕ' to ϕ''' are desired.

$$\phi = (x_4 \wedge ((x_1 \rightarrow x_2) \wedge (\bar{x}_3 \Leftrightarrow x_1)))$$

⇔⇔⇔ V/D/TY ⇔⇔⇔

School of Computer Science and Engineering

Winter Semester 2022-2023 Final Assessment Test

SLOT: G1

Programme Name & Branch : B.Tech CSE

Course Name & code: Design and Analysis of Algorithms BCSE204L

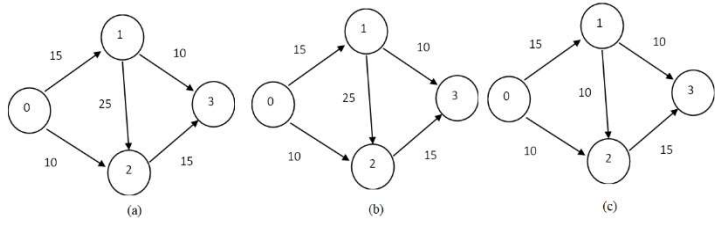
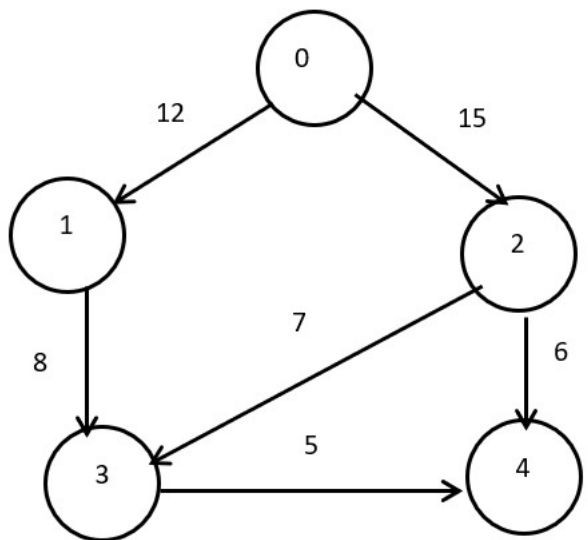
Class Number (s):

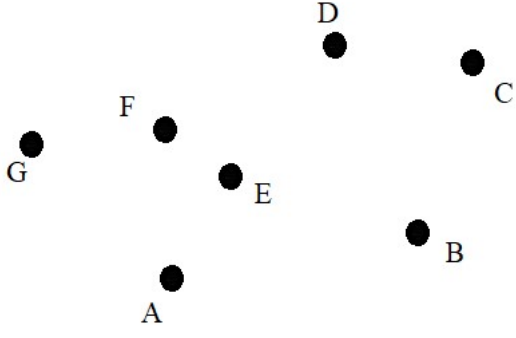
Exam Duration: 180 Min.

Maximum Marks:100

Q.No.	Question	Max Marks	CO	BL
1.	How does Karatsuba algorithm handle numbers with odd number of digits? Write the algorithm/program and show the stepwise execution of the same to multiply the two binary numbers – 10101 and 11. Analyse its time complexity.	10	CO1	BL3
2.	a) There are two knapsacks – one with a capacity of 8 and the weights and values of the items that can be used for filling it are (3,4,6,5) and (20,30,10,40) respectively and the other with a capacity of 20 and the weights and values of the items that can be used for filling it are (15,10,10) and (9,5,5) respectively. Validate the correctness of the statement “Greedy approach does not ensure an optimal solution for 0-1 Knapsack problem” using the given examples [5M] b) Two different character encoding schemes E1 and E2 encode the characters R, G, B, Y as (0, 011,111,11) and (0,11,101,100) respectively. The string “RGRBYRR” is encoded using both the methods, transmitted and decoded in the receiver side. Which of the two is Huffman encoding? Justify your answer. [5M]	10	CO1	BL5
3	Given below is the pseudocode to implement Dynamic Programming technique for matrix chain multiplication. Explain the reasons for assigning these specific initial and final values for the ‘i’ loop and ‘k’ loop and the ‘j’ variable.		CO2	BL2

	Matrix-Chain(p, n) <pre>{ for ($i = 1$ to n) $m[i, i] = 0$; for ($l = 2$ to n) { for ($i = 1$ to $n - l + 1$) { $j = i + l - 1$; $m[i, j] = \infty$; for ($k = i$ to $j - 1$) { $q = m[i, k] + m[k + 1, j] + p[i - 1] * p[k] * p[j]$; if ($q < m[i, j]$) { $m[i, j] = q$; $s[i, j] = k$; } } } } }</pre> return m and s;																																							
	Apply the same to fill the table given below and find the total number of scalar multiplications required to be done to multiply the following 5 matrices of ranks [4,10,3,12,20,7]. [5M+5M] <table><tr><td></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>0</td><td>0</td><td>120</td><td>264</td><td></td><td></td></tr><tr><td>1</td><td></td><td>0</td><td>360</td><td>1320</td><td></td></tr><tr><td>2</td><td></td><td></td><td>0</td><td>720</td><td>1140</td></tr><tr><td>3</td><td></td><td></td><td></td><td>0</td><td>1680</td></tr><tr><td>4</td><td></td><td></td><td></td><td></td><td>0</td></tr></table>		0	1	2	3	4	0	0	120	264			1		0	360	1320		2			0	720	1140	3				0	1680	4					0			
	0	1	2	3	4																																			
0	0	120	264																																					
1		0	360	1320																																				
2			0	720	1140																																			
3				0	1680																																			
4					0																																			
4	Write the code snippet / pseudocode for implementing backtracking technique to find the subset of elements from a given set, whose sum adds up to a given target value. Employ the same to find the first two subsets from the set [10,20,50,60,80] that sum up to the target 80.		CO2	BL3																																				
5	Construct the Longest Prefix Suffix (LPS) table for the pattern “121231213”. What is the length of the longest proper prefix and suffix that can be extracted from this pattern? What is the length of the proper prefix-proper suffix that can be extracted from the pattern “121231”? If the characters from indexes 0 to 6 of the pattern (“12123121”) match with the text and the 7 th character does not match, which character in the pattern should the next comparison start from? Compare its time complexity with that of Naïve pattern-matching algorithm.	10	CO3	BL3																																				
6.	“To find the maxflow using Ford Fulkerson algorithm, it is required to add the path flow along the reverse edges	10	CO4	BL4																																				

	also in the residual graph, as there may be a need to send the flow in the reverse direction, sometimes”. Following certain heuristics for the Ford Fulkerson algorithm, the first augmented path selected for the given three flow networks (a,b,c) are (0-1-3), (0-1-2-3) and (0-1-2-3) respectively. Proceed to calculate the maxflow for the 3 networks and determine which of the three scenarios help to justify the aforementioned statement. Consider source=0 and sink=3 			
7.	Apply Push-Relabel algorithm to find the maxflow for the graph given below. Source=0 and sink=4. 	10	CO4	BL3
8.	a) From the following set of line segments, find all the pairs that intersect, using a more efficient method than the Brute Force approach. [5M] {(2,8),(6,9)}, {(5,7),(10,7)}, {(3,5),(8,4)}, {(7,6),(12,12)}, {(9,10),(16,3)}, {(14,7),(17,9)} b) Apply Graham's scan algorithm to find the convex hull of the following set of points. [5M]	10	CO5	BL3

				
9.	Design a randomized algorithm for the hiring problem and perform probabilistic analysis to evaluate the computational complexity of the algorithm.	10	CO6	BL2
10.	Brief the various complexity classes and illustrate the conversion of SAT problem to 3-CNF-SAT problem.	10	CO7	BL1



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Final Assessment Test – May 2024

Course: BCSE304L - Theory of Computation

Class NBR(s): 0758/ 0762/0764/0767/0769/0770/

0773/0783/0788/0794/0842/0859/1011/1013/1024/ Slot: B1+TB1

1027/1028/1031/1034/1038/1040

Time: Three Hours

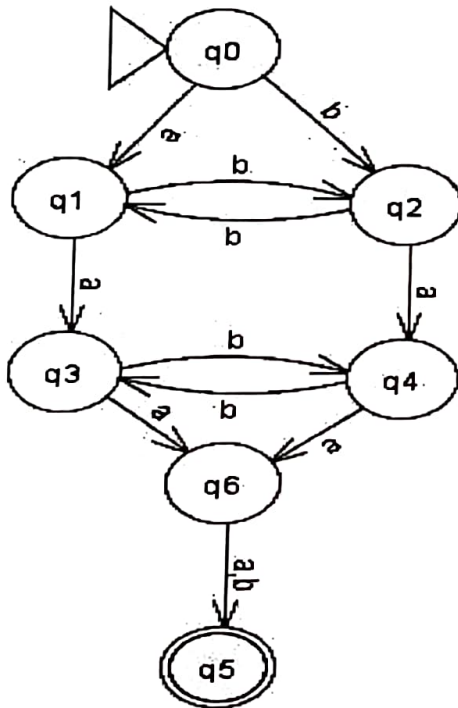
Max. Marks: 100

- KEEPING MOBILE PHONE/ELECTRONIC DEVICES EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer any TEN Questions

(10 X 10 = 100 Marks)

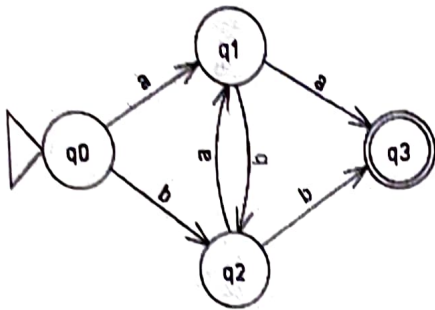
1. a. If $x+y$ is an even number, then prove that x and y are of same parity [4]
(even-even or odd-odd) using contra-positive.
- b. $L_1 = a^n b^m$ and $L_2 = b^m$, where $n, m = 2$ find $(L_1 \cap L_2)^2$ and $(L_1 \cup L_2)^2$ [6]
2. Minimize the following DFA using table filling method



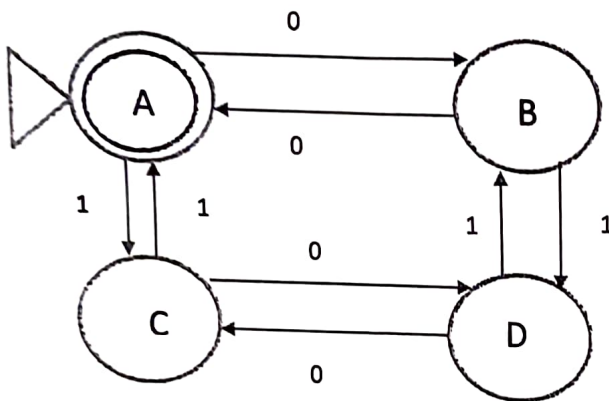
3. Convert the following NFA with epsilon moves to an equivalent NFA without epsilon moves where the starting state is q_0 and final state is q_2 .

States/input	ϵ	a	b	c
q0	Φ	{q0}	{q1}	{q2}
q1	{q0}	{q1}	{q2}	Φ
q2	{q1}	{q2}	Φ	{q0}

4. Derive the regular expression from the below DFA using Ardens and State elimination method.



5. a) Convert the given finite state automata accepting the language $L = \{w \mid n_0(w) \bmod 2 = 0 \text{ and } n_1(w) \bmod 2 = 0\}$ where $n_0(w)$ denotes the number of 0's in w and $n_1(w)$ denotes the number 1's in w to an equivalent regular grammar. Where the start state is A. [5]



- b) By using pumping lemma, prove that the following language is not regular [5]
 $L = \{a^n b^l a^k \mid n=l \text{ or } l \neq k\}$
6. a) Prove the context free Grammar, G is having ambiguity for the string $w = aaaabbbb$, (S is the start symbol). [5]
 $S \rightarrow ASB \mid BSA \mid AB \mid BA$
 $A \rightarrow Aa \mid a$
 $B \rightarrow Bb \mid b$
- b) Convert the given context free grammar G , into an equivalent context free grammar G' in Chomsky Normal Form (CNF), where the start symbol is A . [5]
 $A \rightarrow BAZ$
 $B \rightarrow aBA \mid a \mid \epsilon$
 $Z \rightarrow AbA \mid B \mid bb$

7. Check the membership for the string $w = babab$ by using CYK algorithm for the given grammar, where S is the start symbol.
 $S \rightarrow AC \mid CB$
 $A \rightarrow CA \mid a$
 $B \rightarrow AB \mid b$
 $C \rightarrow CB \mid b$
8. Construct a PDA for the given language $L = \{a^n b^{n+m} c^m \mid n, m \geq 1\}$ and show the moves made by the PDA for the following strings (a) $aabbbbcc$ (b) abc .
9. Construct a Turing machine for the language $L = \{a^n b^n c^n \mid n \geq 1\}$ and show the ID of the Turing machine of the string from abc to $a^2 b^2 c^2$.
10. Discuss any four types of TMs focusing on the transition function with a sample transition rule. Draw necessary diagrams at the required places.
11. Discuss
 - (i) Any two closure properties where recursive languages are closed
 - (ii) Any two closure properties where recursively enumerable languages are not closed.
12. Define Chomsky hierarchy of grammars with suitable examples and their corresponding machines.

$\Leftrightarrow \Leftrightarrow \Leftrightarrow H/E/TX \Leftrightarrow \Leftrightarrow \Leftrightarrow$



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Final Assessment Test – May 2024

Course: BCSE304L - Theory of Computation
Class NBR(s): 0760/0763/0766/0768/0771/0774/
0784/0789/0797/0845/0861/1014/1015/1026/
1029/1036/1037/1042/1043/1045/1047/1049/
1051

Slot: B2+TB2

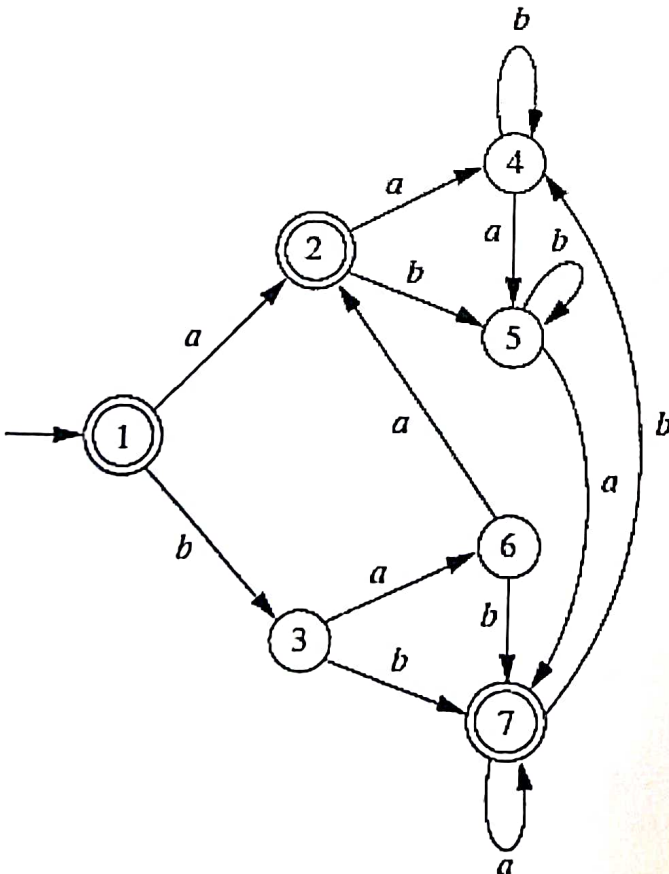
Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ELECTRONIC DEVICES EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer any TEN Questions
(10 X 10 = 100 Marks)

1. a) Construct a Context Free Grammar for the following languages: [6]
 - i) $L = \{ a^m b^n \mid n = 2m, m, n \geq 1 \}$
 - ii) $L = \{ a^n b^m c^k \mid k = |n-m|, n, m \geq 1 \}$
- b) Use induction on n to show that $|u^n| = n |u|$ for all strings u and all n . [4]
2. a) Construct NFA over the alphabet $\Sigma = \{a, b\}$ which accepts all strings whose length is divisible by 5. [5]
- b) For the constructed NFA in Question 2a, construct an equivalent DFA. [5]
3. Construct a Minimized DFA using Table Filling Method



4. Consider the regular expression

$$R = (aa)^* + ba^* + (ab)^*$$

- a) Construct an NFA for the above regular expression. [5]
b) Construct a regular grammar G, for the language generated by the regular expression $aa^* (ab/a)^*$. [5]

5. Construct a DFA for the following language over $\{a,b\}$.

$$L = \{w \mid w \text{ does not contain substring } aa \text{ or } bb\}.$$

From the obtained DFA, use any appropriate method and derive the corresponding regular expression.

6. Determine whether the string $w = \text{He eats a fish with a fork}$ is in the language generated by the given grammar G, using the CYK algorithm.

$G = \{$
 $S \rightarrow NP VP$
 $VP \rightarrow VP PP \mid V NP \mid \text{eats}$
 $PP \rightarrow P NP$
 $NP \rightarrow Det N \mid \text{He}$
 $V \rightarrow \text{eats}$
 $P \rightarrow \text{with}$
 $N \rightarrow \text{fish} \mid \text{fork}$
 $Det \rightarrow a$
 $\}$

7. a) Find an equivalent grammar in CNF for the given grammar. [6]

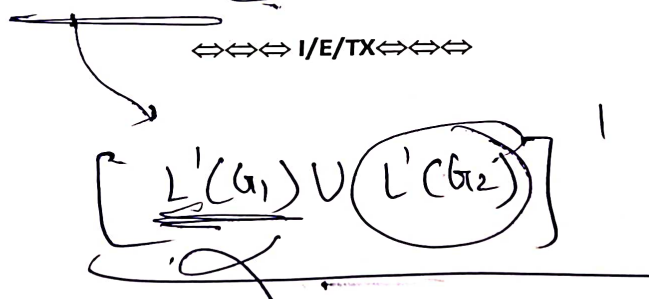
$$G = \{(S, A, B), (a, b), (S \rightarrow ASB \mid \epsilon, A \rightarrow aAS \mid a, B \rightarrow SbS \mid A \mid bb), S\}$$

- b) Check whether the following grammar is ambiguous or not. Show the derivation steps. K is the start symbol. [4]

$$K \rightarrow QK \mid \epsilon, Q \rightarrow Qa \mid aQb \mid ab$$

8. Construct a Pushdown Automata (PDA) that accepts the language $L = \{a^n b^{m+n} c^m : n \geq 0, m \geq 1\}$ by empty stack method. Give the Instantaneous description for the string $aaabbbbcc$.

9. Design a Turing Machine transducer that takes the input $w \in \{a,b\}^*$ and produces a copy of it as an output. For example, if $w=aba$, then the output will be $abaaba$.
10. a) For an Universal Turing machine, obtain the code for $\langle M, 1010 \rangle$ where $M = (\{q_1, q_2, q_3\}, \{0,1\}, \{0, 1, B\}, \delta, q, B, \{q_2\})$ have moves
- $$\begin{aligned}\delta(q_1, 1) &= (q_3, 0, R) \\ \delta(q_3, 0) &= (q_1, 1, R) \\ \delta(q_3, 1) &= (q_2, 0, R) \\ \delta(q_3, B) &= (q_3, 1, L)\end{aligned}$$
- b) Does PCP with two lists $x=\{b, bab3, ba\}$ and $y=\{b3, ba, a\}$ have a solution? Justify. [6]
11. Construct a Turing machine that will accept the language $L = \{a^n b^m a^{n+m} : n \geq 0, m \geq 1\}$ over the alphabet $\Sigma = \{a, b\}$.
12. a) Discuss recursive languages and recursively enumerable languages. [4]
 b) Let G_1 be any unrestricted grammar, and G_2 be any regular grammar. Show that the problem $L(G_1) \cap L(G_2) = \emptyset$ is undecidable. [6]





- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
➤ DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions
(10 X 10 = 100 Marks)

1.
 - a) Let $L_1 = \{ n^2 \% 3 = 0 \text{ and } n \in \mathbb{Z} \}$, $L_2 = \{ n \% 3 = 0 \text{ and } n \in \mathbb{Z} \}$. (Note: % is modulo operator). Prove by contrapositive, "If L_1 then L_2 ". [6]
 - b) Let $\Sigma = \{ \#, \$, \& \}$ be an alphabet. How many distinct strings of length 5 can be formed over Σ . [2]
 - c) Let $\Sigma = \{ a, b \}$ and consider the language [2]

$$L_1 = \{ w \in \Sigma^* \mid w \text{ begins with } a \text{ and } |w| \geq 3 \}$$

$$L_1 - L_2 = \{ aba, abb \}$$
 Determine one possible definition for the language L_2 (using set builder notation or formal description) such that $L_1 - L_2$ is satisfied.
2.
 - a) Write a Regular Expression for a language that cannot have two consecutive a's together over $\Sigma = \{ a, b \}$ [3]
 - b) Based on the above regular expression which you obtained in the Question No. 2. (a), convert it into Non-Deterministic Finite Automata. [7]
3. Convert the following Context Free Grammar (CFG) into an equivalent Greibach Normal Form (GNF). Also prove the string "abab" is derivable in 4 steps of Right Most Derivation via the resultant GNF.

$$S \rightarrow SS \mid aSb \mid ab$$

S is the start symbol of the CFG.
4. Design a Push Down Automata (PDA) for

$$L = \{ a^i b^j c^k d^l \mid i=j \text{ or } j=l, i, j, k, l > 0 \}$$
 and justify whether your obtained PDA is deterministic (DPDA) or non-deterministic (NPDA)?

5. Design a Turing Machine (TM) that performs the following operation over $\Sigma = \{0,1\}$. Also briefly discuss about your idea for constructing transition rules and explain how you process example 3 or example 4.

Copy Paste operation

For Example 1 – Input : 0011

Output : 00110011

Example 2 – Input: 010010

Output: 010010010010

Example 3 – Input: 00

Output: 0000

Example 4 – Input: 111

Output : 111111

6. Either Prove or Disprove the following statements:

[3+4
+3]

- If L and $\bar{L}$ are Recursively Enumerable, then both languages are recursive
- If L is recursive, then $\bar{L}$ is also recursive.
- If L and $\bar{L}$ are Recursive, then both languages are recursively Enumerable.

7. a) A student X derives a string (w) of length K from Chomsky Normal Form (CNF). An another student Y derives the same string (w) of length K via Greibach Normal Form (GNF). Who would derive the string w quickly? X or Y? Justify. [3]

- b) Let $L = \{w.w^R \mid w \in \{a,b\}^+\}$ is a context Free Language (CFL). (Note: w^R means reversal of w). Answer the following with proper justification. [7]

- If you complement L , will $\bar{L}$ be also a CFL?
- If you reverse L , will L^R be not a CFL?
- If you perform Kleene closure over L , will L^* be not a CFL?

8. a) Let us say a problem $P1$ is given to you. You need to find whether $P1$ is decidable or not. How do you find it? Suppose the given problem $P1$ is an instance of below-mentioned Post Correspondence Problem (PCP), what is your stand in it? [6]

List A: 1, 10111, 10 and List B : 111,10,0

- b) Draw Chomskian Hierarchy and interpret the relationship among grammars, languages and automata from the diagram. [4]

9.a)

Design a Turing Machine TM that computes

$$f(m,n) = \begin{cases} m - n & \text{if } m > n \\ \text{otherwise blank} \end{cases}$$

Where m, n are integers over unary representation of $\Sigma = \{0\}$.

Also compute Instantaneous Description (ID) for $f(0000,00)$ and $f(00,0000)$.

OR

9.b)

Design a Turing Machine for unary to ternary (Base 3) convertor.

For Example 1 - Input: 00000

Output : 12

Example 2 - Input : 000

Output : 10

Example 3 - Input : 0000000000000000

Output: 120

10.a)

Let $L1 = \{ w \mid \text{length of } w \text{ is not even} \}$ and $L2 = \{ w \mid \text{length of } w \text{ is divisible by } 3 \}$ over $\Sigma = \{0,1\}$. Let $M1$ and $M2$ are the corresponding Deterministic Finite Automata (DFA) for $L1$ and $L2$.

Find

- $M1 \times M2$ (means cross product of $M1$ and $M2$. That means DFA accepting $L1 \cap L2$).
- Complementation of $M1 \times M2$.
- Reversal of $M1 \times M2$.

OR

10.b)

- Let $M = (Q, \Sigma, \delta, q_0, F)$ be a Deterministic Finite Automata (DFA) [5]
Statement 1 : If you reverse M , the result automata (M^R) is either DFA or Non-Deterministic Finite Automata (NFA)

Statement 2: If you complement a DFA, the resultant language is also complemented.

Statement 3: If you complement a NFA, the resultant language need to be complemented always.

Analyse the truthness of Statement 1, 2 and 3 and justify your answer.

- Let us assume the length of a string is n . If so, how many states are required for minimized DFA that accepts the following: [5]

- Exactly n length strings
- Atleast n length strings
- Atmost n length strings

⇔⇔⇔ F/E/TY ⇔⇔⇔